

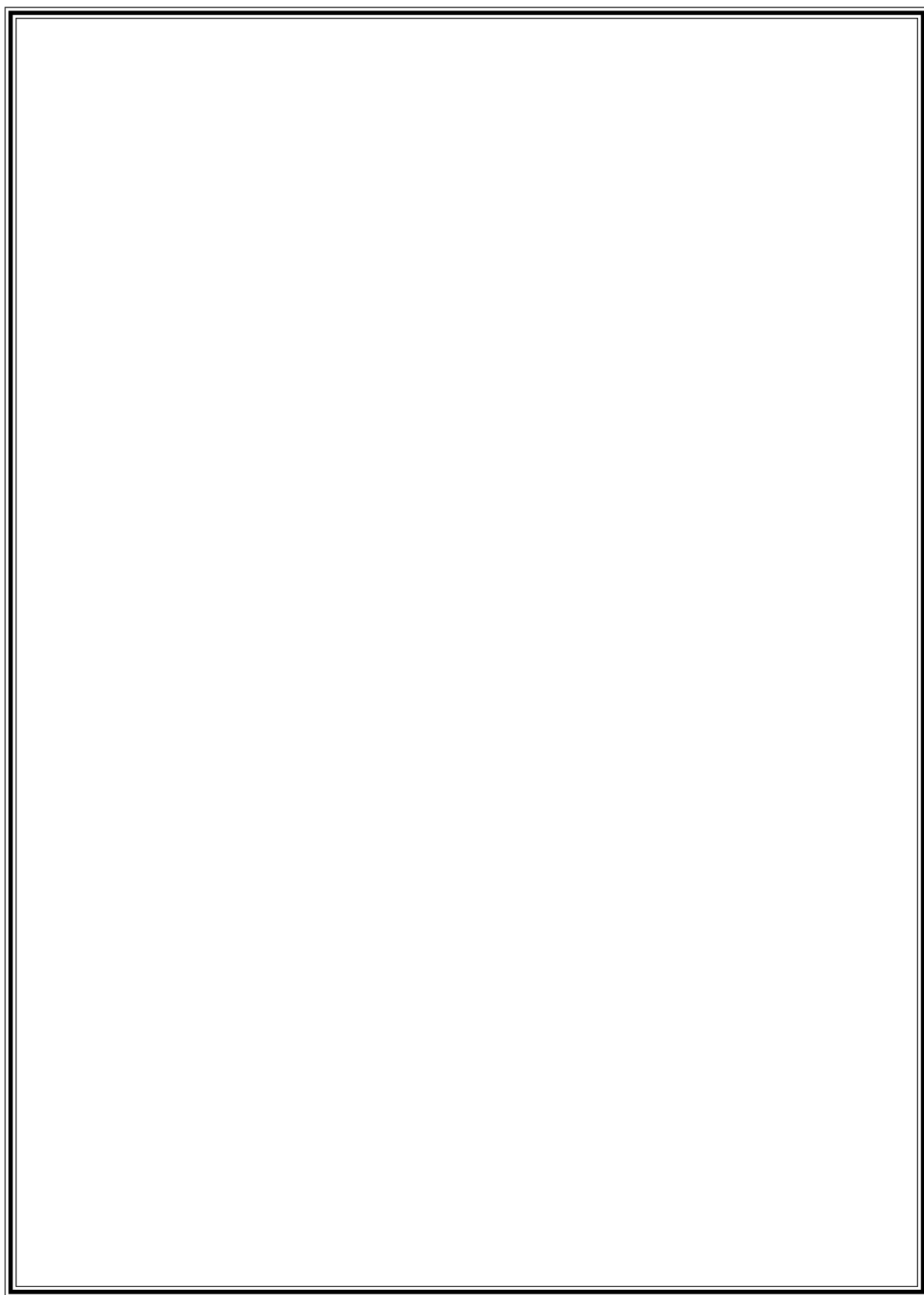
Deploma in Elementary Education

(D.El.Ed.)

First Year

PAPER - VI

ICT



CHAPTER – I

Pedagogy – Concepts and Perspectives

Children come to school with of cultural, social and economic diversity. The teachers leave to be tracked in a way that they understand this diversity

– National Curriculum Frame Work – 2005

The Concepts discussed in this chapter :

- 1.1 The concept of 'learning'. The abilities of children before they come to school. Their applications.
- 1.2 Learning, Learners, instruction, curriculum, school, knowledge – understanding the relations.
- 1.3 To discuss in detail the concepts lies child centred education, activity based learning, joyful learning, CCE, IQ and Merit.
- 1.4 Various methods of teaching and learning – Critical study, providing learning facilities – Teachers as reflective practioners – Cooperative and participatory learning.
- 1.5 Inclusive Education – Concept, Gender, Underprivileged classes – children with special needs.

Objectives :

1. Understand that they should understand the children's nature and teach them to suit to their needs.
2. Understand that children have a prior knowledge of lessons (???) before coming to school.
3. Be able to explain the interrelations among learning, knowledge, teaching and curriculum and schools.
4. Get command over the concepts like child centred Education, Activity Based Learning. Joyful Learning Continuous Comprehensive Evaluation (CCE). Intelligence Quotient (IQ) and merit.
5. Be able to study various teaching learning methods critically.
6. Explain the role of the teacher as a facilitator, and reflective practitioner.
7. Have an understanding on the need of inclusive education and classes that need to be focussed.

Introduction :

'Construction of knowledge' is the first step in the complete development of learner. Teachers have to develop and provide activities that are varied and appropriate to children. Teacher have strive continuously in order to come and of monotonous teaching and to design natural and dynamic learning experiences.

In this process, teacher and pupil both should be active. Teacher help pupils develop and in this process develops himself. The teachers have to make every effort to progress the concept of 'learn to teach' to the concept of 'teach to learn'.

- 1.1 The concept of learning, nature of children, Abilities of children before coming to school – Their applications.

'Learning' is the attainment of new knowledge. According to constructivist school of thought, Learning in the process involving 'construction of knowledge'. The learners construct their own knowledge from the material exhibited to them and the concepts embedded in the activities by adding new concepts to the concepts. For example, the learners from concepts pertaining to transport system if the text book, pictures, sceneries they have seen belong to transport system.

1.1.1 : Learning

As 'A permanent change in behavioural pattern defines learning because of reinforced practice. If this definition is closely examined, it is clear that there are three elements in it.

- This is a behavioural change – It can be progressive or regressive.
- This change is in result to practice.
- This change is almost permanent.

According to Skinner, the behavioural change that leads to the path of progress is learning.

Reinforced practice is the basis for learning. There may be some changes without practice. For example, in the changes that take place because of maturity, there may not be any practice. Growing high, gaining weight, jumping, running, frogs swimming and birds flying etc belong to this category. These changes are not treated as cases of 'learning'.

There is a cause and effect for learning. If the reinforced practice becomes the permanent change, it becomes the effect.

Reinforced Practice

Learning

Permanent change.

1.1.2 : Nature of children in Primary Stage :

Our aim is complete development of child. If we want to achieve this development among children, we need to design suitable activities and programmes. In order to design such activities and programmes, we have to understand children thoroughly. We have to be aware of their interest. If the children's interests are not taken into account while designing the programmes, the children don't take any interest in school and as a result, they move away from school. Let us observe the nature of children in primary stage. Many observations, researches, and opinions made us understand the nature of children in primary school stage as follows :

Nature of children :

- Children evince interest in learning new things.
- They are curious in order to understand new things.
- They want to do. They don't believe anything blindly.
- They are stubborn. They love freedom.
- They want to do things in the way they like to do.
- They don't like to sit at the same place for long.
- They don't want to do the same thing for long. They want to change in activity.
- They are not willing to speak to strangers.
- They like to work with peers. There is a sense of competition among them.
- They don't like fault finding in their activities. If someone criticizes them, they don't want to do the same thing again.
- If we prohibit them from doing something, they try to do it more. They have no fear of doing it.
- They show interest in playing, singing and stories. They love to make dolls.
- They like different colours and colouring something.
- They love to play with soil and sand. They never leave it when they are asked to.

- They are selfish.
- They are highly creative.
- They readily point out to faculty. They correct their own mistakes.
- They love to complain about others.
- They accept challenges. They readily get involved in competitions arguments and discussions.
- There is a lot of self-respect among children.

All the above can be treated as the nature of children.

1.1.3 The ability of children before coming to school – Their implications.

Discussion Point : Are children's minds empty slates ? Discuss.

Ever since the children are born, they would have learnt many things unconsciously by the virtue of their innate talents like observation and thinking. Places like family, environment, neighbours, relations, friends, nature, T.V., Computer, mobile and other such things become the reasons for this. The children learn more as they go to pre-primary Education Centres between 3 – 5 years. Balwadis, Anganwadi centres, L.K.G, U.K.G offer education to children with the help of playing and singing. So, children are equipped with prior-knowledge about many concepts and things before they enter primary school. Let us observe them below.

We can observe the following abilities by the time they enter class I.

- They have understanding of members of the family, neighbours, friends companions and such other issues.
- Regular activities, routine, food habits.
- have knowledge of day and night, time, earth (ground), the sun, the moon and other concepts.
- Listening to stories narrating stories and responding to others in conversations.
- Convey their views to others.
- Speak with error free sentences, pronouncing the words clearly.
- Identify letters and numbers, Reading pictures.
- Know the concepts more-less, depth and height, mass, shapes and constructions.
- Understand concrete concepts.
- Know concepts like water, plants, trees, animals, body, hygiene, good and bad, shelter and clothes.
- Understand the concepts like discipline, conduct, behaviour, playing games, following rules and songs and active songs etc.

When we observe the above points, we know pretty well that children have a prior knowledge of language, mathematics, and environmental sciences. Apart from this knowledge, they also a prior knowledge of health and physical education, work, life skilly, value education etc. If the teachers design activities based on this prior knowledge learning is facilitated and that leads to improvement of children's abilities.

1.1.4 : Classroom applications : The following principles of teaching are to be kept inmind to make teaching suitable to children's abilities. We should teach :

- from known to unknown concepts.

- easy things first and then the difficult concepts.
- introduce abstract things from concrete things.
- proceed from specific to generalisation.
- proceed from whole to the parts.
- move from actual elements to representative elements.
- proceed from deductive method to inductive method.
- teach from synthesis to analysis.

Teachers should use their discretion and decide on the use of above principles while teaching.

1.2.1 : The Learner :

An individual who is in the pursuit of learning is called learner. In school education, child in the learner. Teaching needs to be designed keeping the child's nature. Only then the child gets exposed to knowledge. It is a general opinion that in class rooms the children's opinions don't find expression. They children often listens to the teacher. The activity of the learners is limited to responding to the teachers' questions and repeat whatever the teacher utters.

They can not work on their own or create opportunities 10 on their own. For this reason, curriculum has to internalize the emotions and expressions of boys and girls, arouse enthusiasm among them, encourage their words, make the learners question, help them carry out researches, and share their experiences with others. Their abilities should not be limited to reproduce their knowledge but they should be enabled to link their experience with others. Keeping this in view, There is every need to redesign curriculum.

1.1.2 : Teaching :

The Modern process of teaching in a tripolar process. It can be depicted as follows :

Fig.

In this process the teacher and the learner are linked with the content knowledge.

Teaching is imparting content knowledge :

The learner does not have knowledge in many content areas. He does not know 'all of the content'. Hence, the teacher has to equip the learner with the required information and the content.

Teaching means facilitating learning : The teacher has to create an environment that promotes learning. She has to stimulate interest among the learners to learn.

Teaching is giving guidance : The teacher has to help and give necessary guidance to the learner and facilitate learning.

Teaching is understanding the surroundings : Teaching also implies making the child understand the surroundings and make necessary adjustments with the surroundings and be able to survive.

1.2.3 : Knowledge and Understanding : What to teach the child is a question. The question takes root from a moor deeper question. The deeper question is what are the aims (Objectives of education). If answer for the question is looked from a social, political and cultural perspective, the acquisition of values and skills is important to every individual. It is not an objective. It is a

collection of some objectives the content or the topics selected for this should help achieve all these objectives the content should be balanced and comprehensive. The foundations to knowledge should be laid by the curriculum that provides experiences by keeping in view the development of the ability to think rationally, understand the world through different view points the development of aesthetic sense, working for the economic development and taking part with others to achieve this.

The sum total of all experiences that comprises all thought processes configuration of concepts can be treated as knowledge with the help of this, we can understand the world around us. We can recognise the activities or the total of programmes or the physical ability associated with thought process as knowledge. It helps by to become dynamic is the world and create things. In the process of evolution human beings helped the create knowledge by becoming pools of knowledge themselves, by thinking and feeling and by involving activities. All learners have to create this knowledge for themselves. Then only they will become an integrative. Hence that it is our responsibility to see that part of the knowledge should focus on the learning process, and make the learners a part in the construction of knowledge.

Children should be able to be critically question the knowledge that they come across. The knowledge that they encounter may be from textbooks with bias, it may be from the resources of the literature available in the surrounding. Children cultivate the critical attitude by comparing this knowledge with their experiences, by interpreting it in their own manner. Some women, dalit activity are using songs as a powerful medinery for discussions, interpretations and analysis. Knowledge is in different forms. It can be in the form television programmes advertisements, sings, paintings and such other forms. The learners can make use of these programmes and make use of these thoroughly for discussion among themselves.

The inequalities based on gender, class, caste, world relations are never personal. Neither are these established collectively by a few people. The inequalities and power centres have a very close relation between them. So, we have to be very cautions while teaching these concepts. Different strategies are needed to teach these concepts to different learners. So, the teachers have to provide opportunity of the learners for free expression and facilitate interaction among these ideas. The teachers have to listen to diverse expression of children and not merely exercise the moral authority. They should respect the ideas expressed. The learners should feel secure enough that the grade differences will not impede their natural right to expression.

1.2.4 Curriculum – Concept : The word curriculum is derived from the Latin wordl 'curare'. The word means 'The path away of charist, or the course of running or the means to reach a goal. The collection of a group of topics or learning experiences or table is called curriculum. The syllables or the sum total of concepts or topics to be taught is a part of the curriculum. The sum total of all learning experiences implemented in educational institutions is called curriculum. The curriculum is implemented by the teacher and she tries to achieve the objectives of teaching and learning by implementing the curriculum.

Carter. V. Good, an educationist felt : "Curriculum is the total of set of general activities that are related to different concepts to be provided by the school to help the learner lead a meaning ful in the society. "The Mudaliar commission (1952-53) expressed. "Curriculum is not mere academic concepts. It is the total of all kinds of learning experience that the child is exposed to

these experiences should be provided to learners with the help of different activities in the class, outside the class, library, laboratory, fields, playfields in formal and nonformal means. These experiences promote all development of children and balanced personalities.

Curriculum – The Components : Curriculum is the totality of all learning experiences provided to learners. The experiences include different aspects. These include languages, sciences, arts and co-curricular activities. Apart from these, activities like self study, academic discussions, seminars, symposives, field trips, sports man spirit, important national and international events, work experience, social usefulla ctivities, creative writing, observation, experimentation and other diverse activities can also be considered a part of the curriculum

1.2.5 The School :

The concept of school in the modern age is broader in sense. John Dewey, an environment educationist called school a miniature society. A scientist by name S.K. Saideen (spelling) called school a people's school. He was of the opinion that the functions of school should be based on the needs and problems of people. Such schools are called community school. There are many community schools in philippines, Denmark, England and USA. The community establishes schools for the sale of its children. It provides everything that is needed to establish school. In turn, the community use the resources and accommodation of school whenever needed.

Group Discussion : Based on the above, discuss the functions of schools in groups.

When we take different opinions of different educationists, the main functions of school are as follows.

Functions of the School :

- 1) **Conservation of Culture :** The school should transmit culture that is invited by to another generation. The school should also cultivate cultural values among the children to enable the child lead a happy and comfortable life as far as possible.
 - 2) **Promotion of culture :** It is the function of the school to transmit the inherited culture from one generation to the other. Conservation of culture in a part of functions of school. Apart from this, the school should also introduce the new culture to children. With the help of this the old culture gets modernised. It inspires the children by giving a new touch to the old culture through the intellectual pursuits. They the school promotes cultured development through these creative actions.
 - 3) **Allround Development of Children :** The school helps in the all round development of the child. It adopts strategies for the physical, mental, social, spiritual developmehnt of children.
 - 4) **Preparation for Life :** The school makes the individual ready for life. The school provides the knowledge, understanding, skills and attitudes needed for future life. It makes the individual ready to take up different responsibilities to be shouldere in life.
 - 5) **Developing social efficiency :** The school makes the prospective citizens ready in a direction of upholding democracy countributing to the progress of the country while enjoying the rights and shouldering the responsibilities.
- ii) **The School Plant :** "The special surrounding that helps the children who are just liberated from the direct influence of elders grow up in the lines of certain principles is school" – Maria Montessori

That means the school environment that has all faculty for the child is called 'School plant'.

The school plant is a comprehensive concept. It is a complex with the school ground, building,

play ground, furnitures the teaching learning material library, laboratory. In a school plant along with the present needs the facilities that are required in future also will be included.

The school is a children's home when the children spend time in schools. After that it is a public place. It is welfare laboratory meant to promote learning through activities.

1.3 Child Centred Education, Activity Based Learning, joyful learning, continuous comprehensive Evaluation, Intelligence Quotient and Merit.

1.3.1 Child Centred Education

The concept of placing children the most important place is called child centred education. In the traditional system of education teacher occupies the central place. In the modern system of education, the child becomes focal poing. The psychological researches prove the need to give importance to the child.

In child centred education the innate abilities of a child are identified. Educationists like Roussean, Froebal, Montessori proposed that the child be given freedom. The teacher has to estimatge the psychological levels of children, and their abilities and the individual differences among them. The teaching learning process should be inaccordance with them.

Fig.

1.3.2 Activity Based Learning :

In general, all children have certain innate abilities. They are :

- | | |
|--------------------------------|----------------------------|
| 1) Observing | 2) Recognising |
| 3) Communicating (Information) | 4) Describing |
| 5) Comparing | 6) Recalling |
| 7) Classifying | 8) Contrasting |
| 9) Generalizing | 10) Establishing relations |
| 11) Estimating | 12) Questioning |

The children will participate in the activities if they are given activities that ae based on their past experiences. Working with peers, discussing, arguing, experimenting, solving puzzles and such other activities help us in creating Natural Learning Experience or Natural Learning Environment.

Activity means work. Learning by working or learning through work is called activity based learning. Children can carry out these activities individually or as groups or as a whole class activity. Andhra Pradesh Primary Education Project (APPEP) has given importance to activity based learning.

There is no learning without activity – Jean Piaget

The activities that provide Natural Learning Environment should have the following features.

- Activity should stimulate childs thinking process.

- Process create enthusiasm
- Allow everyone to take part.
- be challenging. The children should do them for their own sake.
- be child / grade (class) level appropriate / objective oriented.
- be flexible to allow changes
- Be easy, simple and have multipurpose
- Be useful to multi grade teaching, and be based on the past experiences.
- allow cater to Multiple intelligences (MI)

Natural Learning Experiences with the above features can only lead to learning. But sheer work will not lead to learning. The six principles suggested in Andhra Pradesh Primary Education Project are helpful for child centred Activity based teaching. They are :

- Designing teacher generated learning activities.
- Improving Learning through activities, exploration, experimentation.
- Developing Individual, Group and whole class activities.
- Providing learning experiences to cater to the individual differences.
- Utilising surroundings.
- Making attractive class room attractive by displaying children's works.

1.3.3 Joyful Learning :

Activity : Prepare list of activities to keep children happy in the class room.

Who are our schools for ? Who are the school activities for ? If we question like this we will come to know that our schools are for children, every activity in the school is meant for children. Children spend most of their time in schools. We know the nature of children. They children only do what they really like. Hence, learning should be in accordance with the child's nature.

The innate abilities of children are the reasons for the behavioural pattern of children. Hence, we should try to provide a Natural Learning Environment to bring the best out of children. Only then the children will satisfied and that will lead to happiness of children. Hence, learning becomes easy. The abilities of children develop to the fullest only when the children are happy and satisfied. The teachers should be able to develop activities that develop the abilities of children. Children can learn well if with the help of games, songs, stories and involving in activities. The lecture method should be done away with.

'Playway' method is useful for joyful learning. 'Games' provide freedom and experiences. In the 'kinder garten' method suggested by Froebel and Montessor' joyful learning takes place. Apart from that, action songs, dramatisation, role play, preparation of songs, dramatisation, role play, preparation of models, project works and other such activities motivate the children, and help learning take place non-consciously.

The teacher has prepare a list of indentified topics, lessons and design activities that promote joyful learning. We have to keep the material ready for conducting such activities. We have to plan group as well as individual activities and give clear instructions to children. Later, the teacher should restrict himself to the role of a facilitator and conduct the activities only with children, He can extend suggestives only when it is really needed. Only then there will be a fear free environment

there will be scope for 'joyful learning'.

1.3.4 Continuous Comprehensive Evaluation (CCE) :

We continuously come across the three words – continuous, comprehensive, and Evaluation.

Evaluation : Evaluation is a process that helps the teachers to check how far the children were able to understand.

Continuous : Evaluation is traditionally done at the time of unit tests, terminal exams. But it is done within the school, and outside the school on a continuous basis. The learning gaps will be identified and remedial measures are taken.

Comprehensive : Comprehensive means the development in physical, psychological, moral and cognitive areas (all round development). That means languages, maths, sciences and social studies arts work experiences, values, health, sports and life skills are given the importance.

Continuous Comprehensive Evaluation (CCE) : It involves the evaluation of curricular and co-curricular areas and all round development of children on a continuous basis. Growth, development, interests, attitudes, values; abilities and all such aspects are taken into account and evaluation is done on a continuous basis.

Learn more about continuous Comprehensive Evaluation in the second unit.

Intelligence Quotient – IQ :

We rate children as less intelligent, more intelligent etc. What is intelligence ? How does it develop ? What is the role of intelligence in learning ? and some other relevant concepts had been studied in paper - I.

The ratio of mental age and chronological age is called 'Intelligence'. If this ratio is multiplied with 100, we get Intelligence Quotient (IQ)

$$\text{Intelligence Quotient (IQ)} = \frac{\text{MA}}{\text{CA}} \times 100$$

If the IQ is

Less than 50 – Imbeciles

50 – 70 – Feeble minded

70 – 90 –

90 – 110 –

110 – 120 –

120 – 140 –

140 above

Example : Chronological age of a boy is 10 years. This Mental Age is 8 years. Then the $\text{IQ} = \frac{\text{MA}}{\text{CA}} \times 100$
 $= \frac{8}{10} \times 100 = 80$ (Hence the boy could be dull)

1.3.5 Merit :

The 'Merit' is given to the individual or thing with desirable abilities. It is a common practice that certain individuals, organisations and Governments give away certificates to recognise 'merit'. Students also are given certificates based on the merit.

Eg :

1) SSC, Intermediate, degree examination – Certificates are awarded to successful candidates

2) DEECET, EAMCET, NEET, DSC, TET – Rank cards.

In the past, recruitment to jobs was done based on educational qualifications.

As the competition is more in job market, now eligibility tests are conducted both for admissions and employment. Selections are based on the ranks. From this we know that general qualification is not sufficient, the candidates with more abilities can only get a position of choice.

It is very important to make students aware that along with Intelligence, hard work also is needed to achieve our goals in life. The intelligence of the students help them understand better in less time, and acquire life skills needed for effective better living. It is only when the required content knowledge is provided to them and when Natural learning environment is created, the when Natural Learning Environment is created, the intelligence will help them progress. It has been consistently observed that many intelligent children from rural back ground are not able to fare well in state level national level eligibility test. The main reason for this is lack of guidance, encouragement and proper environment for learning. In another aspect, some children with learning. In another aspect, some children with urban background are not that good in the matters of synthesis and analysis. As a result, they are not able achieve the desired levels of learning. It is our responsibility to see that learning happens and concepts are understood properly. Otherwise, children will forget concepts immediately after the examinations are over.

1.4. Different methods of teaching – Critical study; providing facilities for learning; Teachers as Reflective practitioners, cooperative – participatory learning.

1.4.1 Teaching Learning Process : Teaching and learning are not two separate entities in reality. These two are intertwined with each other. It is an age old belief that the child learns only if the teacher teaches. The child is not static to take what ever is uttered by the teacher. Learner is the key factor in the process of education. Learning is a continuous process.

The individuals keeps on learning from birth to death. This learning is education. There are two factors involved in it. One is teaching, the other, learning. The child is teaching, the other, learning, The child first learns from mother, then from father and later from the other members of the family, peer groups, religious institutions and environment. This learning is spontaneous and non formal. Formal teaching takes place at school. This kind of teaching takes place through teacher. As we have seen earlier, learning is understanding the concepts in teaching and applying the same after understanding the concept.

There are two main methods in teaching learning process.

1) Traditional Method 2) Modern Method.

Let us understand them.

A) Traditional Method : In this method, the teacher's role is important. The decision with respect to the questions like what to teach ? How to teach ? is always vested with the teacher. The teacher decides everything in the process of education. The teachers believe that all children have similar abilities and sense learning pace. So, the teacher expects the same response from all children. But some children are slow in learning, and as a result are subject to punishment in this method. The teaching learning process is a one way process in this method.

Fig.

B) Modern Method : The child is an independent being in the modern method. According to psychology, there are individual differences among learners. This is quite natural. The child's mind is not a blank slate. The child is endowed with certain abilities to grow mentally and physically. Educationist like Montessori and Froebel, there is a lot of things are hidden in child's mind. The teacher has to make the children think and bring out the creative abilities from the children. Simply put, teaching learning process is to make children think.

This thinking is a process. It needs a purpose and a motivational force. This process that stimulates force is known as teaching. According to the modern method, the process of education should stimulate creative abilities and its creative thinking. The differences between creative thinking and discriminative thinking are as follows.

Creative Thinking	Discriminative thinking
– It is exploratory in nature.	– It is rational nature
– It is motivational	– It is to test imagination
– It is spontaneous	– It is purposeful.
– It is adventurous in nature	– It is limited
– It is decentralised	– It is centralized

1.4.2 Different teaching learning methods – A critical observation.

Though the objectives of education and curriculum are sound, if the child does not understand, she becomes a challenge to the education system. The teacher can only place him/her on the path of understanding. The teaching method an effective can act as tool in this regard. This teaching method can be decided taking 3 factors – the learner, the objectives of education and curriculum. The objectives and plans are based on the thoughts of philosophers. It is philosophers. The decision with regard to make education accessible and making the thoughts functional are decoded by philosophy.

In the olden days teacher centred education was important, now the child centred education is in vogue – Today we should tailor our teaching keeping child in view. The Idealists practised question answer method and discussion method. On the other hand, realists relied on experimental method, correlation method, exploration enquiry method. The Naturalists were of the opinion that learning is possible through actions and self experiences. Pragmatists advocated the use of topical coordination and correlation through utilitarian process.

Let us have a close look at different types of teaching methods :

- 1) Question – Answer Method :** In this method, the teacher teaches by eliciting answers from students to questions. This method is also called Socrates method. The discussion method which is popular today is based on the this (Socrates method) with minor modifications.
- 2) Inductive – Deductive Method :** This method was practiced by Aristotle who was a disciple to Plato. The process of arriving at a general concept from a specific concept is called deductive method. The process of explaining a specific concept in the light of a general concept is called Inductive Method. This method is very much useful in sciences is very much suitable research activities.

- 3) **Self Activity Method** : Pestalouai was of the opinion that the child can learn a lot through self activities. he argued that teaching in at all fields should be base on psychology. He proposed that teaching based on the personal experiences will help in the acquisition of permanent knowledge. This concept led to adopt 'play way method'.
- 4) **Training Method** : Herbert was of the spinion that training, and education belong to the same concept. Training is the basis for education. There is no education without training. But training which is different from education is not desirable. Stuffing the child's mind with unnecessary information can not treated as training. Training should help purification of mind in order to provide the information to the learner. The teacher should be patient enough and train the learner sympathetically. While giving required suggestions. This is not possible in the absence of patience. The undivided differences among learners will create certain obstacles. Hence, the teacher should be ready to take special interest on the learners.
- 5) **Play way method** : Play is a pleasant experience to children. They express their instincts, desires through playing. They come into contact with all things in nature and go into a state of extreme happiness. This happiness leads to learning among children. The essence of naturalism is to teach lessons with the help of play way method. This was perceived by some educationists like montassori and froebel and advocated the method like 'kindergarten'.
- 6) **Learning through Activities** : When children involve themselves in activities on their own, they understand so many new things. Techniques and values are learnt on their own and the knowledge thus acquired is stored in brain. So, they will be able to understand the right concepts fast. This method has so much of importance in teaching activities. Based on this principle, children are engaged in certain activities and opportunities are provided to learning by doing.
- 7) **Making Hypothesis** : When the children take up an activity and organise it, there will be learning in making proposals. They select the topic and decide on how to conduct the activity. Similarly, they learn.
- 8) **Experimental Method** : The main feature of this is learning through 'effort'. When the learner makes an effort on his own. Learning will be fast. There is no use if we sit idle. There will be some awareness when they do something. So, the environment needed for experiments will be created and provided to them. The learner learn as they work on it the experiment.
- 9) **Problem solving method** : The learners can solve the problem that they face. John Dewey advocated that the child be involved in useful, creative and problematic activities. In this process, the internal and external energies get ready for learning. They utilise the entire surroundings and body and engage themselves in learning.
- 10) **Project Method** : Project method is a boon given to the field of education. The teacher suggests some problems and make the children find out the possible solutions on their own. The learners participate in various constructive activities and achieve natural result.

What ever be the teaching method, the positive results will come only if the teachers try to bring awareness among the learners while utilising the available resources according to the topic / lesson.

Activity :

- 1.4.3 **Facilitating Learning** : If we want the children to learn well, the class room environment should facilitate children's learning. The teaching - learning process also should be in accordance with

that. For this to happen. The teaching learning process followed by the teacher should promote constructive learning environment.

Present situation :

- The children's participation is very less in the present learning environment.
- The teaching learning processes are mechanical.
- Children are sitting only at one place. There is no room for discussion among themselves.
- There is no room for the children to question or to share their experiences with teachers on the peers.
- The teachers are not creating tasks that give a chance for the children to think.
- There is no place for new trends in teaching.
- The children are not able to continue to learn meaning fully, as there are no attractive teaching learning processes.

What should we do as teachers ?

We should design meaningful teaching learning activities which give scope to children's participation. For this, we should act not only as a teacher but also a facilitator in the classroom.

The characteristics of a facilitator :

A facilitator :

- Give more importance to listening to others than speaking to others.
- appreciates whole heartedly
- gives more importance to maintain cordial relations than getting the task completed.
- gives importance to know to the others view point rather than expressing her / his opinion.
- takes a decision collectively. It is never autocratic.
- responds appropriately to the situation
- possesses skills to use others resources available than using his / her energies and skills.
- looks straight into others eyes confidently while interacting with others.
- is able to explain the rationale to others.
- acts as a helper not as an officer.
- doesn't act like a scientist, acts like a trainer.
- has natural enthusiasm towards human beings, issues and life.

Traditional teacher

Gives importance to objectives and knowledge.

Participates solely in the classroom teachers in the manner he/she likes.

Children are passive and sit at one place fixedly.

Teaching in always in lecture method.

Teacher as Facilitator

– Ones importance to children's experiences.

– creates suitable teaching learning environment.

– At children participate in the teaching learning process actively.

– Teacher – children (whole class / group) learn meaningfully through discussions.

gives sheer information	– helps construct knowledge by the active participation children in teaching learning process in groups / individually.
children listen to the teacher obediently. There is no scope for discussions.	– Makes the children share their experiences in groups or whole class activity.
Importance is given to give answers to questions / the way information is given	– encourages children to face challenges and get diverse experiences.
Very limited opportunity to express opinions. No chance to given to think.	– gives importance to child's opinions and logical thinking. Encourages children to question and answers their doubts.
encourages competition among children.	– gives opportunity to mutual cooperation and team work and promotes learning.
gives importance to children's memory.	– gives importance to creative abilities of children and critical thinking.
gives importance to information knowledge. No score for activities / experiments	– promotes construction of knowledge by participation of all children in teaching learning process and thereby develop.

1.4.4 Teachers as reflective practitioners : Knowledge exchange takes place among and groups of children through experiences and background knowledge. They can get understanding either individually or in groups through interaction. They can progress. They can develop.

They can develop conceptual development because of the rational attitude. They question and know reasons. They have to explain the cause and effect relation of their understanding through discussions. So they develop rationality. The children will be able to construct knowledge with the help of this procedure. The role of teacher is very much essential in bringing about a complete change. The teacher has to be a colearner, lead the discussions and encourage the children to ask enquiry based questions. The teacher have to work with children and make them a part and express the main ideas in teaching learning process through expressions and partnership. Only then the teacher will become a researcher, cooperative learner and more than that a facilitator rather than a guide.

In this process, teachers keep on expanding the scope of their self learning. It has now become prestigious for the teachers to become thinkers and reflective practitioners. They help the children grow and along with the children, they also can grow. The teachers have to raise to the level of 'teach to learn' rather than 'learn to teach'. We have to learn how to learn. Make the children learn. If necessary we should be ready to unlearn what was learned or correct.

1.4.5 Cooperative and participatory learning :

Children learn in diverse ways. They learn through experiences, activities experiments, reading, listening, responding after thinking, reflecting and writing. If all these opportunities are given to children they derive more benefit. In this diverse process children derive benefit individually on

their own and with the help of others. – NCF 2005

Naturally children evince interest to move in groups, learn by doing. Even if we engage them in mechanical teaching learning processes, when they get opportunities to learn by doing, they participate in those learning contexts. There will be scope for them to share this experiences with others, discuss and demonstrate their abilities and skills as they participate in work. So children evince interest in participating in a learning experience.

Can all children learn in the same way ?

Madhu and Gopi started making a kite one day. They did not know how to make a kite. They went a neighbour and asked her to help. She taught them how to make kites. But, the kite that Madhu made did not fly. The one made by Gopi flew successfully. Both of them got training from the same person. They took suggestions from the same person. Both of them tried to make the kites in the same way. But how did one kite fly and the other did not. Let us think for some time why did this happen ?

Both of the children were equally interested in one activity why only one was successful ? Discuss.

Though children participate in learning contexts discuss the process with peers and groups all children can not learn one thing exactly the same way. If one child learns cycling perfectly in two to three days, some other boy may not be able to learn the same skill in a perfect manner. yet another boy may depend on some others to get on to the seat and get off the bicycle. This is a very common observation to all of us. Finally, though everyone learns to ride a bicycle, everyone has some uniqueness. The method of learning will be different. Let us discuss why there is a lot of diversity in this.

- though everyone is interested to learn though there is a need, individual abilities influence learning.
- There are differences in receiving and understanding the instructions.
- Identifying the tools and using the same will be done differently by different people.
- The need and priority of learning the topic will be different for each learner.
- The encouragement and support from friends, peers and parents will influence learning.
- The topic or skill to be learnt may not be sufficiently challenging.
- Lack of opportunity to demonstrate the skills or knowledge that is acquired or learnt.
- The topic learnt may lead to another new topic.
- Though the difficulty level of the content is the same for all, some may feel difficult and some may feel it easy.
- Children may not notice their own abilities and be in state of inferiority complex.

For all the above reasons there will be different paces and patterns of learners among all children. Only if we know that children are different, and they learn in their own style, the children will be able to participate to the full measure in the learning experiences 'provided to the children.

It is possible to adopt suitable teaching strategies and make the children participate in the teaching learning activities only if we agree with the fact that the children learn differently. The learning activities, methods that are developed and are based on the children's process of learning will be worth implementing. They yield good results. The following activities can be created as a

part of cooperative and participatory.

Learning :

- 1) **Experiencing :** Children should take part in the organisation without any body's help. As a result, they gain experiences.

E.g. : – Model making

- Participating in dramatisation
- demonstrating
- problem solving
- participating in games

- 2) **Sharing :** Opportunities should be given to share experiences with others.

- Sharing their observations actions and results with every body.
- Enquiring of their self experiences.
- Discussing the feelings gained through experiences.
- Talking freely about their thoughts individually or in groups.

- 3) **Processing :**

- discussing, analysing and responding
- discussing how experience was gained
- discussing how problems, concepts and ideas emerged out experiences.
- discussing how to solve problems.
- discussing personal experiences
- encouraging team members.

- 4) **Generalising :**

- Linking experiences with real life contexts.
- Understanding processes and truths from experiences.
- Identifying real life principles.
- Identifying the key factors that improve learning.

- 5) **Applying :**

- Applying the knowledge gained from the previous experiences to new contexts.
- Discussing how the knowledge or skill gained can be applied to new contexts.
- Discussing how the knowledge or skill will be useful in future.
- Discussing the change in behaviour through new learning experiences.
- Assisting individuals in getting command over the topics learnt and make the individuals feel the ownership.

Thus the process of learning with mutual cooperation by making the learners a part of the learning process is called 'cooperative participatory learning.

- 1.5 **Inclusive Education :** Concept, Gender, Ill privileged groups & children with special needs.

There is every possibility that the differently abled children are likely to be isolated by normal children, the class rooms should keep the needs of all (suitable to all) and the teaching learning

process be designed. – NCF 2005.

All the children of today are heirs to the next generation. The future of the country depends on the development of children. All children have a right to develop. They have a right to develop without any discrimination on the grounds of region, caste, religion, gender. It is the responsibility of all to provide equal opportunities to all children. In this direction, a development plan that moves forward inclusively is the need of the time. Children will be able to develop without any discrimination only when education acquires the nature of inclusiveness.

What is Inclusive Education ?

Inclusive Education intends bring all the children of today's society under one umbrella. It is aimed at providing equal opportunities to all. That means : Inclusive Education is the process of making quality education accessible to all without any discrimination on the grounds of class, gender, differently abled, children with special needs socially deprived sections and Nomadic tribes who are far away from towns.

1.5.1 Historical Perspective :

'Education for All' in the watch word. All school age children should be in schools. Education should be made accesible to all. With this aim, many types of schools have been established. There are children of different categories like migrating children, children with special needs and children of different backgrounds. Separate schools have been established to provide education to these children. Because of such exclusive schools, children of these schools are deprived of certain opportunities to promote equality extended in general schools (mainstream school). In order to overcome this, there in a need to provide education to all in one school. In order to achieve this goal, Inclusive education has become a reality to include all categories of children. Let us observe different models of inclusive education.

- A) Traditional Model :** The society and the parents were of the opinion that having children with disability is a curse. Treaging this as an insult, parents used move away from the society. They used to keen the children with disability away from the society. As a result, children did not have access to education.
- B) Medical Model – 1 i.e. segregation)** Based on the disability, the effect of the disability was ascertained. The needed medical facility is provided to such children and special arrangement like establishment of special schools was done.
- C) Medical Model – 1 :** Children with special needs (CWSN) have been provided inclusive educatrion in formal schools in the place of special schools.
- D) Social Model : (Inclusion)**

The obstacles for learning in case of each child need to be identified (by children, Government, parents and society) and they need to be addressed by all concerned.
- E) UNESCO Model :** Inclusive education is continuous process increase the participation of all learners in learning, cultural and social activities while addressing their specific disability needs and see that the learners are not deprived of classroom processes and education.
- F) CSIE (Centre for Study on Inclusive Education) Model :** Inclusive education is not just limited to specific groups of children. It encourages learning and participation of all children. All includes the rulers, children, their parents, staff and society.

If we analyse the present condition of the school : Though there is a responsibility of providing uniform school education to all, in practice, the conditions are not found. At present, children from all class are admitted into school, it can be observed that parents and society don't treat the children of weaker sections are not treated as the normal children. They are treated specially. This can be observed from the following examples.

- Though in the children with special needs (CWSN) dropouts, child labour, and children from different weaker sections, they are treated separately. They are not treated on par with others.
- Teachers don't show much interest to these children as the teachers strongly believe that those children can't learn as well as normal children.
- They are under the misconception that the progress of the normal children is severely affected because of the inclusion of the ill privileged children with the normal children.
- The right opportunities are not provided to the ill privileged children as they are not aware of childrights.

Why Inclusive Education ?

- to make children of all classes join schools and to make them continue studies.
- to create confidence among the children with special needs "we can learn".
- to provide quality education to all children without any discrimination against caste, religion and class.
- to provide equal educational opportunities to children who are suffering from HIV/AIDS
- to provide the opportunity of receiving children of normal school in children of nomadic tribes or of weaker sections.
- to enable the children of multi lingual context with out any linguistic discrimination.
- to drive the inferiority complex among children away from the children, and increase their self confidence.
- to promote the values of cooperation working together, having empathy towards others among the children.
- to make children learn in a liberal environment.
- to enable all children to participate in cocurricular activities actively.

Inclusive Education – Features :

- Identification of boys and girls from different languages, castes, religions, classes, inequalities, children sufering from diseases, children with special needs as a part of inclusive education.
- protection of children from punishments, verbal abuse, and discrimination.
- provision of equal learning opportunities to all children from different backgrounds.
- Involvement of family, society and all teachers as partners in the development of learning of children.
- Improvement of children's participation and cooperation in order to coordinate.
- Provision of opportunities for the children to learn in a free environment and develop.
- Encouragement to children to participate in different learning processes and makes children ready for learning.

1.5.2 Inclusive Education – Impediments :

- 1) Gender discrimination
- 2) Language problem
- 3) Caste discrimination
- 4) Disability
- 5) Attitude of the teacher
- 6) Rigid curriculum
- 7) Narrow examination pattern
- 8) Parental attitude towards education.

The above are the impediments to the implementation of inclusive education. Let us discuss the points that need to be focussed in order to overcome these.

1.5.3 Inclusive education – sections / classes to focussed : Generally all children living on one habitation will not have the same social, economic, cultural background. In certain places children certain groups are subjected to discrimination on the grounds of caste, religion, gender. This discrimination adversely affects learning. Those children may drop out from the school. We see physical, mental and emotional diversity within the children of same group. All these children should be provided with inclusive education.

Let us look at the story of a student : There lived a boy named Rahim. He did not join a school till he was 10 years old. In a special drive called 'Badi Baata', teachers admitted Rahim in 4th class. He used to feel inferior as he did not have the needed learning levels. Rahim did not understand the language that the teachers used and lessons and maths lessons. The teacher did not notice this. Rahim did not know how to approach the teacher and tell him that he was not doing well in studies. He took his problem to the notice of his parents. They said that teachers are the right people to solve his problems. They could not help him as they were uneducated. Rahim did not know what to do. The teachers continued teaching. Rahim's problems became more intense. Everyone in the class knew that Rahim was not comfortable. He felt that nobody paid attention to him. In a few days time, Rahim found a mechanic shed.

Let us think :

- What experiences did Rahim face in school ?
- Why did Rahim drop out of school ?
- Will Rahim get legal support ?
- What all reasons are there for children like Rahim drop out of the school.
- What changes are needed to retain children like Rahim ?

Let us observe what kind of children should be focussed to provide Inclusive education.

1) Children who are slow in learning : The teacher is teaching mathematics to class IV children. Ravi, Sarala, Mahesh, Mamata were not able to carry out additions, subtractions and multiplications. As a result, they are not able to carry out divisions. Teacher did not pay attention to these and went on to complete divisions and proceeded to teach fractions. Let us just think can all children actively take part in the 'Fractions' ? Which of these people will be interested in learning fractions ? How far are they taking part in learning ? Isn't there a danger that all these children dropout of school ? So, is there need of inclusive kind of education to these people ?

- 2) **Intelligent children :** The teacher designs activities keeping the average children in view. The children who can complete will try to complete early and they wait for the other children to complete. The way these children behave in idle time is looked at as 'misconduct'. So, these children get discouraged and they move away from other children.
- 3) **The children who are frequently absent :** Some children are frequently absent from classes. They can not participate actively in the learning activities. They don't know what happened in the class the day before.
- 4) **Children with multilingual background :** In some habitations, there are children who speak different languages. Their mother tongue is different from the instructional language or medium of instruction. There are differences in the children speaking the same language. The teacher has to respect their language in the class and make those learners and make them participate in the activity. Cooperative spirit should be established between children of different languages and make them know better about each other.
- 5) **Children from migrating families :** In certain places, children from frequently migrating families are looked down upon. Even teachers don't pay much attention to the children with a view that "these people come only for few days" and hardly show any interest to their note books and examinations etc. As a result these children will not show much interest in learning. Since their participation in learning is discouraged, there is a possibility that the gap between these learners and the school is likely to widen : The problems to these children have to be considered with sympathy and ensure their participation in learning and examinations.
- 6) **Girl :** In many schools, girl activity is limited to only a few activities. There is general "These children can't do" attitude towards girls. An activities experimenting, projects preparation of resorts boys are given priority. As a result, girls feel that they are inferior.
- 7) **Age – Class (Grade) mismatch :** Certain children are forced to learn with children of younger age group. They can not mingle freely with others because of age, maturity etc. They are under the feeling that the teachers and the children of other groups will make fun of them. As a result they can not participate in learning freely. Special attention should be paid to these children to make them participate in learning and encouraged them to drive the sense of inferiority from them.
- 8) **Children with Emotional problems :** In some children, the emotions are not in their control. They tend to cry out for silly reasons. Behaviour of certain children is strange and extraordinary. Other children in strange and extraordinary other children are scared of these children. As a result, nobody comes forward to accept these children as members in their teams in school / class activities. The teacher have make an and make them move freely with others counselling them.
- 9) **Children with special needs :**
 - Children with physical disability.
 - Children with hearing impairment
 - Children with defective vision
 - Children with mental retardation
 - Children with cerebral palsy.
 - Children with learning problems.

Children with learning impairment and defective eyesight can not decode the teacher's written

or oral message. They can not actively participate in group activities. Unless there is a strategic support from teachers and peer group they can not progress in learning. As a result, they may get detached from school. So, Inclusive education is essential to such children.

- 10) First Generation Learners :** First generation learners face the problem of lack of guidance and death of facilities. The peers exercise authority over these children. There will be less parental cooperation and as a result that will impact attendance. These children should be made a part in participations learning giving proper guidance.
- 11) Children with HIV/Health problems :** There are several problems with children with HIV or other severe health problems. It is particularly a problem to sit together, dining etc. There is a chance that they people may drop out from the school as the school programme, peers in learning don't welcome the companionship of this group of children. Teacher has to provide inclusive education to such children.
- 12) Child laboures in school :** There will be some difference in term of age, language, habits of child labourers. Certain children make fun of thes children based on the work the child labourer did or the salary they got. Inclusive education has to be provided to these children by establishing friendly relations with these children.
- 13) Children of Tribes / Nomadic Tribes :** The tribal children or the children of nomadic tribes are different from the other children in terms of traditions and customs language and attire. The teachers should develop strategies to make their influential cultural background respected by other children, and these children take part in all learning activities and all school activities.

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- 3) B.Ed, Elementary education, 2010, Telugu Akademi, AP Hyderabad.
- 4) Inclusive Education

Practice Exercises :

I) Essay Type Questions :

- 1) State various teaching learning methods and, explain the features expected in a natural learning environment.
- 2) What is Inclusive education ? What are all the different groups / sections to be focussed in Inclusive Education.

II) Shor Answer type questions :

- 1) What is the nature of children in primary level ?
- 2) What are the functions of school ?
- 3) Explain the importance of activity based learning.
- 4) State the role of teacher as facilitator.
- 5) What type of activities can be designed for children as a part of cooperative – participatory learning ?

III) Very short answer questions :

- 1) What abilities do children possess before they come into school ?

- 2) What is learning ?
- 3) What are the differences between traditional and modern teaching learning methods ?
- 4) What is curriculum ?
- 5) What is school plant ?
- 6) What are the six principles suggested by APPEP ?
- 7) What is the role of teacher in 'Joyful learning'.
- 8) What continuous comprehensive Evaluation ?
- 9) Are IQ and merit the same ? or not ? How ?
- 10) Explain the role teacher as 'Reflective Practitioner'.
- 11) Is inclusive Education necessary in the present context ? Why ?

IV) Objective Questions ?

- 1) Match the following

A) Learning	()	1)
B) Teaching	()	2) Change in behaviour
C) Montessori	()	3) Guiding
D) Traditional method	()	4) CWSN
E) Inclusive Education	()	5) One way method
	()	6) Lecture method

- 2) The set of all programmes implemented in school is called
- 3) We can consider as a set of experiences as a result of pattern of thoughts or the shapes of ideas.
- 4) One example for innate abilities of children.

CHAPTER – 3

Concept and Processes of Learning

Content of the Chapter

- 3.1 Learning Processes among children – How do Children Learn?
- 3.2 Behaviourism – Fundamental Concepts and Applications.
- 3.3 Concept formation – Meaning of Concept.
Concept formation – Bruner, Piaget.
- 3.4 Thought and discrimination: The relation between thought process and learning
- 3.5 Cognition – Learning.
Cognition Development, Structuralism – class room applications.
- 3.6 Vygotsky's Theory of learning – The concept of ZPD – Classroom applications.
- 3.7 Construction of knowledge – The Role of a Teacher.

Objectives:

- 1. Understand how children learn.
- 2. Explain the fundamental concepts of Behaviourism and classroom applications.
- 3. Meaning of 'concept', The psychological processes involved in concept formation.
- 4. Be able to differentiate the concepts of thought and learning.
- 5. Know about cognition, learning and constructivism and their classroom applications.
- 6. Understand Vygotsky's theory of learning and be able to apply to the classroom situation.
- 7. Know the need of critical pedagogy and classroom applications.

Introduction:

? ? ? (page100

Learning in change in behaviour. This change can come through training. How to design learning processes for children, and how to make changes occurred through learning permanent is an important question to consider. It is expressed that the opinions and theories need to be used depending on the context. It is very important to discuss how to develop concepts, cognition, thoughts and discrimination among children. It is very essential to discuss how to develop concepts apply to the concepts to situation.

3.1 How do children learn?

Children have natural talents. The teacher has identify these talents continuances and tap these abilities and develop. Every child is unique. The teacher will succeed if he/she understands the uniqueness to each child. There are so many misconceptions about the child's concentration, memory etc. As a result of the misconceptions teaching learning process does not progress. Sometimes we don't know whose fault is it for the problems in learning.

Children learn in many ways. Along with self experience, making things, putting the knowledge into practice through experimenting, reading, discussing questioning, listening, thinking, writing, and through such other activities children learn individually or in groups. If we carefully observe child's psychology carefully, every child has his own likes and dislikes, interests, skills conduct and behaviour. With this diversity the child thinks that he/she in a unique being.

This unique behavioural pattern begins with family, society, classroom form childhood and develops.

- Children construct new knowledge based on the prior knowledge they have.
- Children apply the concepts learnt in to school to real life by way of questioning.
- Children respond differently one question by answering differently.
- Children keep thinking in novelty on their own or with the information form media.

- These skills are honed in school, at home and in society in the process of using them and working.
- There are observations, research, questioning, discussing in taking active part.
- By applying to new context proposing new theories, new thoughts, and innovations take place.
- Because of challenging goals children will be able to think independent over the problems in multiple ways.
- Creativity and self discipline develop among children problems in multiple ways.

“People call me a great teacher. In fact, I just made my disciples think. Nothing else”.

----- Socrates

“In the world there are three kinds of people. First kind of people change with their experience. The second kind of people change with others experiences. The third kinds of people never change with their experiences or with other experiences.

-----Chesterfield.

3.2 Behaviourism-fundamental concepts, and their application. With pavlois theory as the base, J.Bwatson , B.F.Skinner and A.L.thorndike proposed Behaviourism learning in the change in behaviour. Behaviour is dynamic. It can be changed with the help of favourable conditions; change in behaviour can be brought about. This is the understanding on which behaviourism was proposed. Conditioned learning, theory, trial and error learning theory problem solving, conditioned learning belong to behaviourism school. It is worth noting that all these experiments were conducted on animals let us look at the solicited features of behaviourism

- Learning is the formation of S-R. That means child learns by the formation of bond between stimulus and response. Stimulus is the input. Response is the output. If we understand this, then there is mother question when the same teacher is teaching, why all children are not learning in the same way.
- Behaviours are dynamic, skinner is of the opinion that behavioural change can be brought about in phases and with proper planning this is rational to some extent. But planning should be done taking the individual needs of the child.
- Are giving home work to children in the lines proposed by Thorndike. No. We are giving the same home work to all children. The be given to slow

learners, and gifted children. If we keep the child in mind, learning should become individualised, as proposed by skinner.

- Behaviourists take the child as a conduct able unit. This understanding leads to mechanised approach of learning. We can motivate the learner to learn with the help of external stimulus. But interest need and recognition contribute a lot to learning.
- Watson once referred to a sensational issue and said, give me healthy children. I will bring them up in suitable environment and make them as you wish them to be. Watson tracts environment as a factor that strongly influences learning. We bring about desirable changes only if we keep the innate abilities of children.

Think! and respond

Discuss how much of behaviourist trends are there in our teaching learning process

3.3.1 The Concept.

Activity: The students are asked to get into groups. Some objects are given to each group and they are asked to classify according their size. Some vegetables are given to children and they are asked to separate them.

Children separate the thing by their quality like big and small.

Concept means

1. It is group of objects having common features expressing
2. The relation between objects, situation, incidents through symbols

In our real life we come across objects size, time distance and such other concrete, abstract concepts. The concrete concepts gradually become abstract concepts. The student who first happens to see school will be able to imagine the school if there is no school that is directly visible.

If proper concepts are not formed with the help of partners and teacher, there is a chance that children will form misconceptions.

3.3.2 Psychological processes like sensitise, perception, generalisation, and intelligence help the formation and development of concepts.

1. Sensations: perceiving the surrounding using sense organs.
2. Perceptions. Meaningful sensations.
3. Generalisation: coming to conclusion using experiences
4. Concrete experiences: Direct experiences
5. Intelligence: ability to think abstractly.

3.3.2.1 Concept formation Bruner.

Bruner discussed the process of concept formation in famous his book. The process of education, proposed the theory of instruction. He was of the opinion that we can teach learners of any age at any time if they are interested to learn (Refer to chapter 2 for Bruner's theory of instruction)

According to Bruner, the concept formation happens as follows:

1. To form any concept the principles of the concepts are to be taught.
2. Easy concepts are introduced first and difficult concepts are introduced later.
3. The concepts are introduced through dramatisation first, then pictures and finally through symbols.

e.g.: To teach addition concept, we can call two children, give five pencils to one and five to another, we can ask them to express the number by joining them second stage in picture form

$$5 + 5$$

Third stage: $5+5=?$

4. Children should be motivated
5. Parents, teachers and students have to explain answers to doubts asked by children
6. First concrete experiences are to be introduced. Then abstract concepts and symbols need to be introduced.
7. Concepts are to be introduced through play way and method.

3.3.3 Piaget – concept formation-

The Piaget thoughts regarding concept formation have led to revolutionary changes. Cognition is the process of understanding

According to Piaget, the concept formation occurs in the following manner.

1. The concepts form became interaction between of experiences
2. The concept develop became of organisation and adaptation
 - a. Organisation: Managing the complex factors carefully with coordination.
 - b. Adaptation: In Adaptation, there are assimilation and adjustment.

The new born child has shows some reflexes. The child chses the eyes on seeing light and holding other person tight became of fear are examples for reflexes. As the child grows old, the child enriches his concepts with the help of experiences and environment

For example form the past experiences the who comes across a donkey student forms a concept that it four legs and a tail. If student happens to see a dog, he thinks it is donkey as it has four legs and a tail. But he differentiates the donkeys bray and the dog's bark

In adaptation, the child uses the change with the help of assimilation when the happens to came across donkey, dog and the other animals they can differentiate their features.

3. Four stage in concept formation:

1. Sensory motor stage
2. Per motor stage
3. Concreter motor stage
4. Abstract motor stage

The concepts are to be introduced depending on the stage and age of the child. First concrete concept are introduced after that, abstract concept are introduced

Bruner, Piaget's concepts help children understand concepts and easily and per mauently.

3.4 Thought reasoning inter relation between thought – learning

The words – thought, discrimination, and learning are heard in daily life in school and classroom context the must. We hear them unconsciously

The right thought and its practice make wounds

Activity: The student are divided into groups the motional leaders, cricketers, film heroes and heroines and some places are given to them. This are asked to speak about them. The students reading speak about the persons they know very well. If they are not known the students will take some time.

Thought is –

1. An internal problem solving nature
2. A cognitive process that forms in a person's psychological domain

Reasoning:

Reasoning is the process of using thoughts in a stages and moving towards goal. Reasoning helps in problem solving

For example, the statements given in bank recruitment tests, railways, navadaya examination for question on logical reasoning

e.g.:

1. Ramu has 2 pens 2 pencils
Ravi has 5 pens 4 pencils
Rakesh has 8 pencils 6 pens

In all, how many pens are there?

2. *****
3. *****

The relation between thought and learning

Learning is an experience there is an almost permanent engage in the individual. That means children continue that learning with experiences.

e.g If we touch hot things, that burns _____ us. If we use vulgar language we will be punished. If we go to school late, we will be asked to stand outside

the change in behaviours is brought about first by mother later with experiences we learn many things with training

e.g: How to conduct prayer, how to sing a rhyme, how to recite a rhyme, ad how to ride a bicycle we learn only through training.

Thought is an internal problem

Solving nature.

Thought / reasoning / solution to problem.

Thought leads to reasoning. In turn that leads to problem solving. Many kinds of thoughts help us problem solving. Integrated thought process leads to better learning (*****)??? Page 106), to think better

3.5 Cognition – learning

Cognition means the development in intelligence or own knowledge of and of the environment or understanding them.

The understands the surrounding with the help of the experience and interaction with the surrounding with its own knowledge. It adjust with surrounding accordingly students develop on the basis of its experiences, knowledge. It done not need any external effort. We should be able to create suitable environment suggestions and guidance is needed depending on the need.

Contrary to S-R learning theory as proposed by behaviourists, Jean Piaget, Vygotsky and Chomsky supported the cognitive school of thought.

Child's mind is not a clean slate (tabula Raga). This basis for cognitive thought.

By the time the child joins the school it has the influence of surrounding and personal experiences. These are called cognitive maps.

- The child constructs new knowledge on the basis of the previous knowledge. Knowledge is constructed on principles assimilation, accommodation and adaptation, as proposed by Piaget.
- Piaget was of the opinion that learning happens in accordance with the sensori motor, concrete operational and formal operational stages.
- Piaget felt that learning should be meaningful. Meaningful learning means the present concepts are learnt meaningful in coordination with the past experiences. The child is motivated to learn when it is kept in the format a challenge
- The reinforcement theory proposed by skinner is opposed by Chomsky from the view point of language learning does not merely happen on the basis of imitation and reinforcement child picks up the responses and uses them in his/her conversations. When the child uses a new word we are surprised we generally wonder how child is using new words that were not taught finally, we come to understand that child has constructed vocabulary
- The behaviourist believe that the teacher is a facilitator who creates teaching learning activities, cognitive partially endorse this, they want the teacher to be a some of inspiration.

- Cognitivist felt that the children should be kept in real problem situations instead of creating problem situation. Doing so, the teacher needs to continue teaching learning process by guiding the children.
- Chomsky and Vygotsky were of the opinion that language learning does not take place by sheer initiation but it happens in a constructivist manner.

3.5.1 Cognitive development:

The topic cognitive development and its education implication have been discussed in paper-I 'childhood, child development and learning'.

3.5.2 Construction :

What is knowledge? How is knowledge constructed? If we understand these questions/topics, then only we can understand the trends. If we believe that the learner accepts knowledge then our attention. If we believe that the child constructs knowledge, we focus on concept and understanding rather than transmission of knowledge. Behaviourism and Bloom's taxonomies have been accepted by the whole world and are continued till now. Those who were strongly influenced by behaviourism focused on how much the children learnt what they learnt, but they did not pay attention to how the child learnt. Gagne (1995), Mayer (1996), Johansson (1991) and Elizabeth Murthy came out with constructivism as a symbol of their thoughts.

If there is proper stimulus, there is proper response. Evaluation should be done on the basis of school processes. Now, evaluation is done on the basis of memory. There is a deeply rooted opinion that there is a need to correct environment if learning does not take place properly. Even we strongly believe that positive reinforcement means improves positive behaviour, while negative reinforcement reduces positive behaviour. Our focus is totally on S-R bond but there is no focus on the process in between. This process is believed to be crucial by constructivists' opinion.

Process	Sciences	Language
	context	context
Video has to be inserted	The learners read the lesson on mammals they watch the video project.	The learners read the story 'catuli wallah' after this, they watch the background in the frame of a story with brief explanation they act out one or two scenes
Observation	The learners make notes of the behaviour of and actions of mammals	The learners watch the films

Contextualisation	The learners analyse the lesson in accordance with the context	The story is applied to the illustrations made of background material.
Cognitive apprenticeship	The teacher gives illustration how the example of mammals can be taken analysing information	On the basis of scene how background material can be applied to story reading
Co-operation	The learners get into group for the conduct of activity and go ahead the guidance given by the teacher	The learners work in groups while interpreting they go ahead as per the direction and guidance if the teacher
Interpretation construction	The learners collect and analyse evidences to establish or prove their hypothesis on the mammals living on earth or in water	They prepare own interpretation on the story

Process	Sciences	Language
	Context	Context
Multiple Interpretations	The put forth their arguments with explanations to their hypotheses or concepts. They analysis the lesson individually and in groups. They try to find answers and inter poet it.	In teams, outside the teams, the learners compare the interpretation they feel that there are many possible relations to 'cabuliwallah' story
Multiple manifest actions	Keeping every context in view , they generalise and several invent in children of mammals	The learners come to know about how the roles in the story can be expressed in different possible ways

We should give importance to the process in the child brain rather than the observable external behaviour mym (1996) compares the teachers to guidance and the learns to constructors of knowledge

3.5.3 Constructivist learning contexts:

Constructivist on lines certain processes of learning they are :

- 1) Observation
- 2) Contextualisation
- 3) Cognitive training

- 4) Co-operation
- 5) Interpretation construction
- 6) Multiple interpretations
- 7) Multiple manifestations

These are called constructivist learning context. during the process of teaching the teacher assists learning , encourages the learners to respond ,analysis ,interest and facilitate the construction of knowledge some constructivist learning context are given here

3.5.3 Constructivist – class room application

The constructivist proposals that paved was to revolutionary thought and practices in education system are as follows:

- The child can only acquire knowledge only when it is absolutely free any pressure if the child in left in a problem situation, the child will learn on its own. There is no need of giving frequent infractions about every minute movement
- Acquisition of knowledge is not main thing. This beliefs, attitude, likes and dislikes help in acquiring knowledge is not the main point for teacher. Belief in profession, having own style in teaching lessons in the important point.
- Social awareness increases by under stating co-operative spirit , cordiality , constructions and sharing . That means the role of socialisation is optical in the construction of knowledge.
- Glaser field (1996) is of the opinion that learning is not just the formation of bond. Self regulation , concepts construction, absorption , responding are a part of learning
- Process of learning is more ilupartant than the product of learning. Learning is continuous process. But how much child the children learn is not important. that means construction is knowledge in primary and not the product of learning
- The aims of education should not be rigid but be flexible
- The present evaluation process intends measurement knowledge and memory of the learner. Evaluation should be a self analysis to the teacher and the learner
- Jonasson (1994) said that learning would take place in the following phases. Make all forms of truth available to child tell the learners the complex truthly about the real world.
- Importance should be given to the result and not to the result
- The various aspects of the content should be made available to the child and allowing the child to make the choice

- Construction of knowledge should take place to cater to the individual needs and within the purview of the social norms
- Vygotsky was of the opinion that the child constructs knowledge in its internal world

3.5.5 Construction of knowledge -5E method

Constructivism is explained by 5E every E of the 5E points to a stage in learning there are:

1. Engage
2. Explore
3. Explain
4. Elaborate
5. Evaluate

This 5E point to the stages of learning new things, continuous evaluations their knowledge and progressing in the path of learning

1. Engage

This is the primary stage .in this stage, the child relates the past experience with the present experience. The learner psychotically prerace to know the new concept

2. Explore :

In this stage the child is provided with a new experience, they develop the concepts, methods, skills in themselves. In this stage, they explore the surroundings and contents thoroughly

3. Explain

The learners question and get the explanation needed. They will be able to demonstrate they skill they acquired recently there is a change in a behaviour. In this stage new vocabulary, definitions and the explanation for the concepts have to be given by the teacher.

4. Elaborate:

This is the stage that elaborates the understanding of the learners. The children will get extensive, deeper understanding through new concepts learnt and collect more information they get new knowledge this stage can be explained as the phases of extensive study.

5. Evaluate :

In this stage, the learners will be evaluates whatever they learn. Teacher also will be able to evaluate their understanding and skills.

3.6 Vygotsky's Theory of learning

Vygotsky is a Russian and he gave a new direction to construction of knowledge through his 'social construction' this book 'thought and language' is famous explaining the process of language learning.

Vygotsky and piaget opposed the behaviourist concept of S-R bonds role in learning. They proposed that construction of knowledge takes place through experience they supported cognitive.

According to constructivism, the individuals construct knowledge through experienced they were of the opinion that learning is nothing but construction of knowledge. the psychological process should be given more importance than the observable external behaviour the essence of this theory in process of learning is more important than the product of learning.

“Construction of knowledge is the process of inviting new things with the help of their earlier knowledge by the children on their own”

The teacher should become active partners in the process of construction of knowledge by children by participating in the learning processes. Children should be provided with natural children will take part in learning process and construct knowledge. Thought and past experience play a very important role in the construction of knowledge.

Construction of knowledge takes place children by way of questioning, relating the concepts learnt in school with outside experience, responding to problems in different angles and expressing opinions and context similar to them. Knowledge is not transmitted from one person to the other. It is constructed knowledge is constructed through psychological process among children.

Individual knowledge is developed through social interacting human being not only interacts with the surroundings, but also changes them as per the need and adjusts with the surroundings

Vygotsky's important contribution to education psychology is that the human intellect develops only with meaningful social cultural activities only. He felt that children take part in social cultural activities and interact with their only their interaction thoughts continuously change and develop.

Children construct their own knowledge development takes place only through learning the development and construction of knowledge takes place in the background of their social, cultural reality language plays an important role in

the cognitive development of children. These are essentials of vygotskys theory we will learn the other aspects in detail.

Reciprocal Teaching:

Through initially this was intended to teach reading to low achieving children, later it was used to teach all subjects, to all children.

In this method, the teacher forms a group of two o four students which is termed as collaborative group. The content of a book is expresses by one after the other through conversations for cognitive strategies called questioning briefing clarify and questioning and proceeding

Generally leaders (teachers) start the conversation with question in the topic. Children respond to the questions. Additional questions will be asked. In case of a teacher briefs it. The learners discuss the whole issue and get the unknown things clarified finally the teacher encouraging the children to predict what is going to happen in the future based on what is indicated in the lesson. In reciprocal teaching the zone of proximal development are created children gradually taken the responsibility to make the lesson understood as a result students learners acquire the skills and psychological process method for learning.

Zone of proximal development (ZPD):

Some students are asked to solve the pubz in the video game. Later after some time, the students are taught how to solve the puzzles by an expert in solving the puzzle. Then the students are asked to observe the difference in solving the puzzle.

This is called zone of proximal development (ZPD) zpd is the difference between the learning without help and learning with the help of a skilled person.

The silent feature of ZPD:

1. Extension of tools to solve and co-operation
2. Learning fast, easily and meaning fully
3. Effort to make learning permanent

Application in the present:

1. The teacher should be a facilitator
2. ZP should be used when needed
3. The student should be given the needed help and co-operation and allow the him/her to learn on his /her own
4. Free environment should be provided
5. Co-operative and organised in learning should be organised in learning

Collaborative learning:

By nature, people would like to spend time with peer group. According to Vygotsky, skilled peers can help other children to progress. In reciprocal learning, children work as small groups to achieve their common goal. The children form attitude by discussing, talking, thinking collectively, questioning, expressing their opinions fearlessly and interacting with others. Thus, they hone their skills.

A student has some capacity to learn on his/her own to some extent. With the help of someone more skilled, the child can learn completely. The difference between learning on one's own and learning completely with the help of a more skilled person is called the zone of proximal development (ZPD).

Another important concept in Vygotsky's theory is scaffolding, which is the extension of help or cooperation in the process of learning.

According to Vygotsky, the gadgets like computers, internet, encyclopaedic entries, video clippings, libraries and library functions as ZPD. Vygotsky called this social scaffolding. The same concept was referred to by Bruner as instructional scaffolding.

Vygotsky classified the psychological process as two there are:

1. Lower order mental functions
2. Higher order mental functions

1. Lower order mental functions :

These are also called as internal capacities. These capacities are natural and inborn e.g.: observing, identifying, comparing, knowing etc.

2. Higher order mental functions :

The interaction of the individual with the society, the lower order mental function in the individual develops to the higher order e.g. Analysing, reasoning, socialising, thinking creatively, reasoning, discriminating etc.

The individual is able to use higher order, mental functions, and knowledge of language and face the complex situations or else they become slaves to the coordination's.

Higher order mental functions control arrogant behaviour, and they help self regulation.

Cognitive development the role of language

Vygotsky was of the opinion that language plays important roles in the cognitive development of children. He believed that children learn, think, converse with others and express ideas with the help of knowledge of language, language development.

With the help of language the child knows that everything in the surroundings, material, social practices will have a name and is useful in some way or other (the language process are used to converse with elders teachers and peers.)

If we observe carefully, the children talk to themselves aloud when they are playing. according to vygotsky , the children talk to themselves for self guidance .as the children grow old their mental abilities develop and they will be able to perform activities silent inner speech

We can observe that children when they are involving in thinking speak to themselves even after growing old. Vygotsky was of the opinion that the private speech in the child hood becomes inner speech because of psychological processes. Language helps children to think, behave and act. So vygotsky behaved that language is the basis for all cognitive process.

Educational Implications:

- The teachers have to understand the social backgrounds of children
- Conditions that facilitate collaborative learning in learning process
- Language development and language knowledge among student should be promoted
- In order to promote construction of knowledge among students learning by rote and direct teaching should be discover aged . in order to consistent , it should be linked to the local context and local context
- We have to make children understand the complex concepts for analysing and synthesising information , thought with sound reasoning should be encouraged

3.7 construction of knowledge –the role of the teacher

Till now construction of knowledge ,forms of knowledge the conditions needed for the construction of knowledge have been discussed constructive learning should facilitative learning contexts that help construction of knowledge the role of the teacher is very important in this the teacher is co learner.

4. Information Communication Technology Introduction

The Technology itself is transformative. It's the School, the pedagogy, that is transformative.

- Tanya Byron

Topics in the Chapter:

- 4.1. The concept of ICT, Tools
- 4.2. Computer Hardware, Internet, Text Documents, Spread Sheets, Presentations
- 4.3. Open Education Resources, Handheld devices, Netiquettes (Etiquettes in the use of Internet)
- 4.4. ICT – National and State Policies.
- 4.5. ICT based learning process – Creation of learning Environment, Educational games.
- 4.6. Self Exercise questions.

Objectives:

- 1. Understand the importance Information and communication Technology in Education.
- 2. Acquire skills like Word, Excel, Power point, Internet needed to use computer in teaching.
- 3. Acquire skills to use resources efficiently through web searching.
- 4. Have an understanding on National and state policies on ICT.
- 5. Be able to manage an ICT class using digital resources.

Introduction:

We are living in an ever expanding digital world. ICT has become a part of our life. We can right from games to social life. We can observe that to days youth is totally relying on online services for everything they need to know. They depend on online services for all kinds of cooperation. Ten year ago, Mobile services were limited to only information. Now use of TV, Mobiles, Tabs and computers has extended and entered the field of instruction along with other fields. And it is providing learning experiences to teachers and learners.

It is very much essential that the trainee teachers should possess skills in handling and ICT class room, keeping the changing society.

4.1. ICT concepts and Implements:

What is information and communication technology? Let us know the concept of ICT and the implement involved in it.

4.1.1.

The expanded form of ICT is information and communication technology. ICT can be used in learning process.

1. What is ICT?

Information and communication technology can be defined as the set of sources and various technical devices is able to expand. We can use many devices in class room for the sake of learning.

4.1.2. ICT devices:

4 type's devices are useful for us in providing entertainment to the children. They are:

TV (Television), Mobile / Tablet, Radio cum Tape Recorder, Computer.

1. Live: They are transmitted straight from the place of incident.

e.g: Independence Day celebrations, Discussions, Conferences

2. Pre – recorded:

Pre – recorded programmes are produced with the help of a pre written script, then they are recorded with a few additions of artistic skills and are made viewable.

The plays with acting.

Various ICT devices and Their Principles of transmission

Televisions (TV)	Mobiles	Radio & Tape	Computer
One way transmission	Two way transmission	One way Transmission	Two way Transmission
Audio & Visual display transmission	Audio – Visual display and text pictures and data exchange	Only audio play	Audio & Visual display and information to be shared.

Transmission through magnetic satellite	Transmission magnetic tapes	Radio Transmission through satellite & tape will be with cable	Transmission through fiber optic internet
Programmes are not controlled by the user	User control & tape is in user control	Radio programmes are not controlled by the user	User control
Need DTH Antenna	Needs service Provider	Inbuilt system for Radio and tapes required for tape recorder.	Needs soft wares & Internet
Recording our voice is not possible but possible But possible with DTH Antenna	Recording our voice is possible	Recording our voice in tape recorder is possible	Recording our voice is possible

4.2 Computer Hard ware (Internet), Text documents, Spread sheets, Present actions:

4.2.1.1: Computer Hard ware:

Definition: Computer is a machine that assists human beings in performing different complex tasks with speed and accuracy. We should knowledge about the internal components.

The characteristics of a computer:

Speed: With help of the information and instruction, Computer gives us the output in and un imaginable speed.

Accuracy: Computer gives us error – free output with the help of instructions given by us.

Strong Capacity (Memory): The information given to the computer can be stored in computer memory. The information stored is expressed in terms of bites. These standards can be expressed as follows.

1 Byte = 8 Bits

1024 Bytes = 1 Kilo Byte (KB)

1024 Kilo Bytes = 1 Mega Byte (MB)

1024 Mega Byte = 1 Giga Byte (GB)

1024 Giga Byte = 1 Tera Byte (TB)

Reliability: The computer is highly reliable. An application of a computer gives the same result any number of time is used.

Multiple Benefits: These computers can be used in multiple aspects in many fields.

User friendliness: With some minimum knowledge in English, a computer can be used by anyone without any special used training

A Computer can work continuously:

The computer can work continuously without any bore done or tiresomeness because it is a machine.

Different Components / Systems of a computer:

There are different components used in a computer. They are as follows:

1. Input Unit
2. Central Processing Unit (CPU)
3. Out Put Unit

Input Unit	CPU - Control unit and Arithmetic, Logical Unit, Memory Unit	Out Put Unit
	Secondary Unit	

Let us know these in more details:

1. **Input Unit:** Key board, Mouse, Light pen, Joysticks, Graphic Pad, Optical Mark Reader (OMR), Magnetic Ink Character Reader (MICR), Optical Character Reader (OCR), Speech Recognition system, Bar Code Reader, Punch Cards, Punch Power Tapes, Touch Pads, Touch Screen, Micro Phone, Web Camera, Magnetic drupelet cetera are the examples for input divides.
2. **Central Processing Unit (CPU):** The unit to which will provide with the information to be processed is called the functioning of control. Logical tasks, Mathematical Processes are performed with the help of Arithmetic Logic Unit (ALU). Storage of

information is the function of memory unit (MU), Along with this, Storage secondary Unit is used to store additional information.

Processing devices are the tools that are used to store the information provided by us, to retrieve and process the information. The processing device like sound card, Micro processors are also a part of CPU.

3. **Out Put Unit:** After the information is processed, the processed information given out to us in the form of output.

The Output Devices: LCD / LED Units or Monitor, Printer, Plotter, Speaker etc. Can be called the output devices.

Some Technical Terminology:

1. File: A film, a video, a song, a programme etc. is/are stored as a file.
2. Folder: 'Folder' or 'Directory' is the collection of different files. We can keep some folders together in another folder.
3. Disc: Files, Folders are used to store information. For example Hard Disc, Pen drive, DVD / CD, Memory Card etc.

Activity: Create a folder with your name in the computer.

4.2.2 Types of Software:

Computer soft wares are of 2 types.

1. Application software.
 2. Operating system software (or) system software's.
1. **Application Software's:** Software programmes that is developed to perform a specific function a computer is known as Application software.
e.g: M.S. Office, VLC Media Player, Win amp etc..
 2. **Operating System:** This helps in the functioning of the computer. Operating system is essential to make the computer work. It is essential to make other programmes (Application software's) work.
e.g.: Windows, Linux, DOS, MAC etc.

We can observe the operating systems DOS, Windows in mobiles, Similarly, we are using MX player, Candy Crush etc., application software's. Mention an app that you know. Explain how it can be used in the classroom.

Activity: Discuss in groups and write how different apps can be used in the class room.

4.2.2 Internet:

The concept of Internet:

The interlinked computers are called Net of Networks. These networks are again integrated to a larger network. This larger network is called Internet. So, Internet is worldwide network of millions of inter linked computers. With the help of this linked network, people across the world are sharing information, news and opinions.

World Wide Web: (WWW)

The phrase World Wide Web denotes the complex of information available on the net. It is simply called “Web” Web resembles a kind of library. There are crores of pages of texts, images, audios and videos. These pages are called web pages. These pages are linked to the Internet and exist in different computers. There are links between webpages. If we click on the letters or pictures of one webpage, they will take us to another webpage.

In order to visit webpages we need Internet browsers like Microsoft Internet explorer and Google chrome.

The difference between Internet & World Wide Web:-

Internet is a large scale network having countless networks. This connects millions computers in the world and enables them to converse with others. Information in crores of pages. This is a part of Internet. This is one method of sharing information or distributing information.

Services offered by web features of Internet:

Activity: What kind of information are we getting from Internet? Discuss. Internet is offering us a wide range of services. By using them in class rooms we can make teaching learning more effective.

1 – E-Mail: Electronic mail is just like a postal service offered to us by computer. With the help of this mails are sent. E.g: Gmail, rediffmail, Hotmail, yahoomail.

Picture Page 127 TM Book.

How to use email?:

If you want to open a Gmail account with Google search engine you have to open the following home page.

- You have to type gmail.com/yahoomail.com/Hotmail or any other website using Internet Explorer or chrome Browser.

- Press “Sign up” to open a new account. Fill in the registration form that appears and create an account.
 - Press “Sign in” button after creating an account type username and password.
 - You can find emails received in “Inbox” and send mails to others with the help of “Compose”. You can send mails received by pressing “forward”.
2. Search Engines: If we don’t know how to trace the information we need, we can get the same with the help of search engines e.g: www.google.com, www.altavista.com, www.yahoo.com etc...

Picture page - 128

3. Chatting: “Chatting” is the process sending messages online to people in far off placed. We can chat with more than one person at a time.
4. News Groups: New groups are forums where people discuss. Sciences and entertainment issues. People can share their opinions, Suggestions, and instructions through these frames across the world. In addition, we can get answers for their questions.
5. Web cam: Web camera helps is hold video conversations online.
e.g: Amazon, Flip kart etc.
6. E- Commerce: With the help of E-commerce facility we can shop online sitting back at home.

4.2.3. M.S. Office: Microsoft firm developed a software called MS office in order to facilitate office work. There are, 4 important programmes of the four, 3 are important. They are:

1. MS Word
2. MS Excel
3. MS Power Point
4. MS Access.

1. **MS Word or word Processor:** We can use this software to prepare question papers, letters, reports, tables, agreements etc.. easily and attractively.
Activity: Discuss how MS Word can be used in teaching.

2. **Starting Word:** Click start and go to “All Programmes”, go to Microsoft and Click on MS Word – 2007. The word opens.

The word Interface has a title bar above, a menu bar, tool bars and rulers. In the Menu bar, there are File, Edit, View, Insert, Format, Tools, Table, Window, Help commands. These options are useful in giving commands to form the functions we want to.

Picture Page No: 129

Creating, Editing a Word Document:

1. Creating a New Document:

- Go to file menu and click on “New”
- Click on blank document on the “Task menu”
- New Blank document is open.

2. Editing text:

- The process of changing text is called Editing.
- To erase a word click at the beginning or ending of the word.
- If the cursor at the beginning we can use ‘delete’ key; If the cursor at the end we can use ‘Backspace’ key to change the text.
- To delete a large chunk of text, select the whole text and click delete or backspace.

3. Save text:

- Click ‘Save As’ on the file menu
- A dialogue box appears.
- Click on the place you want to save the file
- Type New name at file save.

4. Copying the text:

- Select the text we want to copy
- Click ‘Copy’ on the Edit menu

5. Pasting the text:

- We can paste the copied text where ever we want by clicking ‘Past’ in the Edit menu

6. Changing Page setup:

- Click on 'Page set up' in file menu
- A dialogue Box appears.
- Make necessary changes in Margin, Paper
- Click on Ok.

7. Printing:

- Click on 'Print' in File menu
- A dialogue box appears.
- Select Print option at file name
- Click ok, Select All or Current Page or Page numbers.

Selecting: Drag the mouse on to the text or file.

Closing: Click on File – Close / Exit.

3. Formatting text:

The way changing the way text appears or text style is called formatting.

Changing font:

- Click select all in the edit menu
- The whole question paper is selected
- Click on the font in the Format menu
- Select the font in the dialogue box.
- Click 'Ok'

Changing Size, Style and Colour:

- Select Heading
- Click on font in the Format menu
- Make changes in the dialogue box to get the style, colour and size of choice.
- Press 'Ok'

Crating Bulleted and Number test:

- Select the list to be bulleted
- Click on Bullets / Numbering in format menu

- Dialogue Box appears
- Click on Bullet or Number style
- Click ok

Formatting Paragraph:

- Select the Para to be Formatted
- Click Left, Right, Centre or Justify based on the need.

Getting Help:

- We can type in search space of Help menu and get explanation that we need.

Activity: Prepare a report as a team on an activity conducted in your organization, make changes in MS Word and submit the hard copy to your lecturer as an assignment.

II. MS Excel:

We can use Excel to prepare marks memos, to calculate total marks, and average ranks easily; and to mathematical tables and graphs of an individual's monthly wages.

What is a Spread Sheet?

It is used to store data, assess and maintain just like Excel.

Excel Terminology:

Cell: The junction of a row and a column is cell

Row: It is a group of horizontal cells. These are called rows.

Column: The group of vertical rows is call a column.

Cell Address: Columns are identified with the help of letters; rows are identified by numbers. If we write number after letter that becomes cell Address.

e.g: A12, B4 etc. (D8 is column D and 8th row)

Starting Excel:

Start – All Programmes – MS Office – MS Excel 2007

Excel Interface: When started, Excel opens with Blank Sheet. It has 3 worksheets as Sheet 1, Sheet 2, Sheet 3 etc.

Creating a New Work Book:

- Click File Menu
- New Work Book Task Pane Appears on Screen
- Click on Blank Work Book.

Fig in Page 133 (TM)

Working with Excel:

- Click on Cell 2 times.
- Correct using Delete, Back Space.
- Click on Enter.

(All options like Save, Cut, Copy, Past, Close are performed as they are done in MS Word).

Using formulas:

- Type for mules in the active cell above.
- First type '=' and then type the formula.

e.g: If we require the sum of some data in B10. Keep the cursor at B10 and type. If we want the sum of B4, B5, B6, B7, B8 and B9 Keep the cursor at B10 and type '=' sub(B4:B9) Press 'Enter in B10. We get the sum of the data immediately.

- The signs +, -, *, / are to be used for addition, subtraction, multiplication and division respectively.

Creating Graphs tool:

- If we want to observe Radha's marks in graph. Select Radha's marks.
- Click on Insert Menu in the tool but
- Select a chart of your choice and click 'Next'.
- Make necessary changes (e.g: Name) and click finish.

Working with functions:

- Click on the Result cell. e.g: E51 (Say the number students who get more than 40 Marks in a group of 50 Students.
- Click on function of in 'Insert menu'
- Select COUNTIF in the dialogue box
- The delete the range (e.g. Marks E1:E50) in the argument box.
- Type >= at criteria.
- Click 'OK'
- We get the particulars of students who got 40 or more marks.

Calculating Average or Auto Sum:

- Place cursor in the cell where result is required.
- Click on Arrow mark by the side of Auto sum.
- Select the mathematical process of choice and click on Average or More functions in the box.
- We get the cell range.

Inserting Rows or Columns:

- Keep, the mouse pointer at 'Row' or 'Column' heading.
- Click on row or column in the insert menu.
- New column or row is added.

Activity: Calculate the average of children's Scores in exams. Draw suitable graphs.

III. M.S. Power Point:

M.S. Power Point is one of the an attractive Presentation of some text in very short time.

Uses:

- Power Point presentation can exhibited to more number of people using a projector.
- We can distribute the same in the form of handout to readiness.
- It can be made attractive using pictures, colours, text, sound clip or animate objects.
- It can be changed from time to time.
- We can prepare lessons also.

Starting power point:

Start – All programmes – MS Office – MS Power Point.

Fig in page 135 at TM

Menus in Power Point:

There are File, Edit, View, Insert, Format, Tools, Slide Show, Window, Help, Menus. Which clicked on different commands in different menus, power point works.

Creating a presentation:

- Click on New in the file menu.
- Click on blank presentation on the task pane that appears.
- We get a single blank slide. Every slide has a different layout.
e.g: Title slide, Title slide & text, Title and 2 – column text; Title, text and clipart, blank, Title and table, Title and content etc.
- Type 'Title' in the 'Click to Add title' in the title slide.
- After typing the 'Text' Click somewhere outside.
- Type your name near 'Sub title'
- Click on New slide on 'Insert Menu'
- We get 'Slide layouts'. Select the slide you need.
e.g: Text layout / Title and Text
- Type the necessary information in 'Slide'.
- When 'Enter' is pressed, the insertion point (Cursor) to next bullet.
- Save the Presentation.

Fig – Page 136 TM

Changing the background of the slide:

- Click on Background on Format Menu.
- Click on the colour you need in the dialogue box
- Add more colour and make it attractive.
- Click Apply or Apply to all. All slides will have the same background colour.

Changing the Background picture:

Format menu – Background – Back ground fill list – fill – effects – select – picture – Insert – Ok.

Change the colour scheme:

Format – Slide design – Colour schemes – click.

Inserting the Pictures:

- Select Text and ClipArt slide.
- If we click two times, the available pictures appear in computer.
- We can drag the picture into slide and engage the slide.

Slide sorter: The option helps as arranging the slides in an order.

Slide Show: We can click on slide show and view all slides in an order.

Activity: Select a lesson in Telugu / Urdu and prepare a power point presentation adding audio/ video to make it a multimedia teaching. Project it in the class. Discuss what more changes are needed.

4.3. Open Educational Resources: (OERS)**Hand held devices; Netiquette.****4.3.1. OERS:**

Provide Inclusive Education by creating contexts – NCF 2005.

Teachers have to create constructive, learning situations to teach. It is our responsibilities to pool up of resources that help us in teaching. These resources should be relevant to topics. They should be easily available and suitable to be changed.

The text book caters to most of our needs. But it cannot be a complete resource by itself. Though audio – visual resources are also available to some extent they are expensive.

Open Educational resources are considered tearing resources. They are digital resources available free of cost. These are available in many forms. Texts, audio – visual forms are available for these resources. E.g. Course material, e – books, software's etc.

In the year 2001, Open Educational sources came into existence with Wikipedia across the world. Later, Massachusetts Institute of technology brought out a pool of resources for numerous courses. They are accessible worldwide for free. Organizations like IGNOV, NPTEL are organizing many courses through open Educational Resources (OERs)

Advantages of OERs:

- The learning Resources can be accessible for free. They can be reused any number of times.
- Resources can be altered to suit to needs. They can be easily enhanced depending on the need.
- These resources can be linked with one another or can be used for the creation of new resources.
- These resources thus created can be made available to all. This is facility is called 5Rs. They are: Retain, Re – use, Re vise, Re – mix and Re – distribute.

Some OER websites:

- www.kjanacademy.org; the predesigned lessons are available at this site.
- The websites that offer free course material; www.nroen.metastusio.org, www.tessafrika.org, www.courseara.org, www.nptel.iitk.ac.in.
- Websites that help teachers: www.karnatakaeducation.org. (KOER)
- Website that creates resources with the help of partners. www.wikipedia.org.
- Some OERs, PhET simulations; X mind software etc.

4.3.2. Hand – held devices:

The devices that can be held by hand and are suitable to use. The small sized electronic devices like mobiles, tablets, iPod's etc. can be cited as examples for these devices.

In the beginning, these were made to link to computers, but now efforts are made to use in the place of computers with the help of these apart from information communication, teaching can be done easily. We can derive all advantages that are possible by using web through Internet. With the help of handheld devices with powerful battery, much storage and memory video, audio, multimedia lessons can be taught by teachers more effectively.

E.g: floods, cyclones, plants, Rotation of earth, revolution of earth, forests, wars, The structure of human body – Its parts etc.. can be taught using suitable videos.

4.3.3. Netiquette:

Think Over: What happens if we don't follow net etiquette.

We have to follow certain traditions while using Internet or face book, Twitter, Whatsapp, G-mail, Yahoo mail, Google plus, linked In and such other blogs and social media websites we should follow certain manners. This is called 'netiquette' (Net etiquette)

While chatting or engaging in other conversions, we should not be having in an impolite and indecent way. We should not respond in an appropriate way. Giving false information, involving in illegal activities is not expected. We should not anchorage others to do such things. Action will take according to the cyber laws in force against who reveal or steal others personal information, used photos software's and information.

We should be brief in communication through Internet. We should be polite towards others, we should share any information only after verifying its genuineness. We should respect others privacy. Simple language should be used. All these are a part of netiquette. We should not share spans or respond to such information excessively. There should be a heading to whatever we send. The information should also be brief.

4.4. National and State policies related to ICT:

4.4.1: National ICT Policy at the national level: 2012) NPICTSE – 2012.

The following are the salient features of policy to ICT in high school Education that was announced by the Union Government. (ICTSE – 2012).

Vision:

It should be the vision to prepare youth competitive enough as constructors of a knowledge society for the sustainable social economic development.

Mission:

Utilization of ICT resources and activities to improve accessibility, Quality and efficiency in school system.

Salient features:

- Creation of environment suitable for ICT.
- Preparing an ICT Community that facilitates national building through ICT literacy.
- Accessibility of ICT devices equally to all free of cost. Encouragement to teachers and students to use these resources.
- Exchange of locally improvised digital resources with others by teachers and students.
- Linking of teachers, schools and expert for the improvement of skills in teaching profession.
- Evaluation of the needed research's, experiments to use ICT efficiently in school education.
- Development of Critical understanding 31 about the advantages, limitations and dangers linked with ICT.

4.4.2 ICT – Policy of AP Government salient Features:

- Attempting to get world recognition to the state be achieving health and education, and quality in life.
- Creation of employment opportunities by developing IT Education and agriculture through Good Governance.
- Designing education by changing syllabus, curriculum and courses to suit to the changing technology.
- Development of skills of tutors in teaching with the help of ICT.
- Utilization of computers supplied in CAL, ICT @ Schools, Class (Computer Literacy & Studies in school) and promotion of computer Assisted learning (CAL).
- Initiation of measures to improve the efficiency of class room instruction utilizing digital resources.
- Utilization of ICT for Academic Monitoring.
- Giving instructions to trainers and teachers through video conferences and teleconference.
- Assessment of progress continuously with the help of ICT through teacher, pupil Assessment indicators.
- Improvement of Accountability and transparencies.

- Making at least one e-literate from every home.

4.5. ICT based Learning experiences – Creation of Learning Environment:

Think Over: What kinds of arrangements are needed for the design of ICT based class room.

ICT based learning experiences are to be provided in School education as per the national policy on ICT. Learning Environment is to be created in Class room by utilizing ICT resources to improve quality and efficiency in teaching.

Various ICT devices like self learning material multimedia material virtual class room, simulations that are relevant to lessons are available. Every device helps in creating class room environment in its own way. The teachers have to identify ICT resources and Choose devices that are suitable to their while doing so, the teachers come competent enough to develop. ICT resources on their own or with the help of others.

- Many schools have EDUSAT terminals or DTH facility with the help of them ManaTV, Doordarshan etc channels lessons are being taught directly and indirectly they should be made accessible 100% to all students.
- TV, radio programmes have to be in aspirated into the children's Time Table and they should be properly made use of.
- Computers have been provided to schools through computer Assisted learning (CAL) Every CAL School is linked to primary Schools with not more than 200 Children, They should be utilized with the help of proper plan and implementation.
- As many as 64 CDs were distributed under CAL scheme. We should know the details of these lessons subject wise and should to children (Now they are available in a set of 12 CDs).
- Learning processes to that give online learning experiences may be created through e0 learning.
- Difficult concepts may be made clear through "You tube" Videos through Internet to every school with the help of Wi-Fi / Broad band.
- Teachers and students can share their opinions with others through 'online forums'. We can clear doubts with the suggestions and instructions of experts.
- Teaching under taken through an online class room is called 'Virtual class room' Through these class rooms video based courses audio and animated courses etc. are made available.

- The power point presentations, videos, audios, graphics etc.. Materials developed by experts can be used by teachers to teach effectively.
- Learning can be improved with the help and cooperation of experts, co-learners or teachers and with the help of websites like Blogs, Twitter, Facebook, Whatsapp, Linked In etc.
- Direct experiences can be provided to children with the help of Smart boards or LCD Projector. In Smart boards there is a white board in the place of LCD screen and it has touch control. There is an arrangement on the white board to directly touch and control the computer.
- Learning can be improved with the help of educational games. E.g: CAL – CDs.
- ICT resources which are also called digital resources, educational??? recordings, web resources, educational soft wares etc. Should be used in teaching.

Educational Games:

“Games” are of many kinds. They are digital games electronic games, video games, computer games online games mobile games, etc. They can be broadly categorized into 3 types. 1. Educational games, 2. Leisure games, 3. Educational Leisure games.

Educational gems are entertaining and educative. They stimulate interest among the learners. They provide challenge and incentives to learners whenever they win Puzzles, small problems, quizzes identifying places with the help of places, reading and responding surroundings and responding and such other processes are the basis for designing these games, these games are mainly designed to make the learners understand the lessons.

As a part of Educational Leisure games, there is interesting way of narration, Narratives with fiction. Some of these contexts can assist conceptual learning of lessons. For example, climate, climatic conditions, maps, computer education etc. Age of Empire, The vice city etc.. can called examples to Educational leisure Games.

4.6. Self Exercise:

1. What is ICT?
2. Explain the ICT devices with examples
3. Explain the different parts of a computer.
4. Write about the process of using E – mail.

5. Jaswanth created a document using MS Word. He wanted to increase the font to 32. The methods to be followed. To change the font are given below But, They are not in an order. What is the correct order?

- I. Click on the font size in the list.
- II. Select the text for changing the font size.
- III. Click on the down arrow in the font size

6. Use MS Office in the computer and make the following necessary changes to the file.

- I. Start word
- II. Type a leave letter in a blank document
- III. Type the title 'Leave Letter' on the top, bold it and underline.
- IV. On the lower side, type the house address on the left.
- V. Select the whole text and change it into Time New Roman font.
- VI. Change the font size to 16 for heading, and make it 12 to the remaining text.

VII. Save the file on the Desktop with the name Leave Letter and close it.

7. Start MS word. Prepare a table of the result for the recently conducted Examination. Use the format given below. Save the file with the title 'Exam Result'

Roll No	Name	English Language	Maths	Social Science	Science	Total
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8. Use MS Office in the computer and make the following changes to the Excel file.

- I. Open MS Excel
- II. Record marks of ten students in Telugu and Mathematics that you conducted in the table given.
- III. Use formula and calculate the total marks in two subjects of every students.
- IV. Calculate the percentage of marks scored for each student.
- V. Bold the heading in the table.
- VI. Make borders to the table.
- VII. Change the name of the worksheet to results.
- VIII. Save the file with the title 'Result' and close it.

9. Make the following changes to power point file using MS Office in your computer.

- I. Open power point. Select Blank layout.
 - II. Use word Art and type welcome. Add your name and Roll number below that.
 - III. Insert new slide and Type 'Plants and Animals'. Insert the pictures of a plant and an arrival on either side of the title.
 - IV. Create five new slides. Add Headings at the top position of slides plants on two slides, Animals on two slides.
 - V. Add as many numbers of plants in the first slides and some pictures of animals in the next two slides.
 - VI. Add their names below the pictures.
 - VII. Use 'Word Art' and type thank you in the last slide.
 - VIII. Design all slides using the background of your choice.
 - IX. Save this file in My Documents with the title 'Plants and Animals'. Present the slide show.
10. Take some examples of open Educational Sources (DERs) and explain how they are useful in the teaching process.
 11. Explain 'netiquette' in the use of Internet.
 12. Mention any one handout (hand held devices) and explain how it is used un the process of teaching and learning.
 13. Write the National and State ICT Policies in School Education.
 14. How can you say that we can teach efficiently with the help of ICT based teaching learning process.
 15. How are Educational Games useful in teaching?

Reference Books copy form 144.

5. Integration of information communication technology

“ Technology is just a tool in terms of getting the kinds working together and motivating them, the teacher is the most important “

--Bill Gates

After reading the chapter the student learns

5.1: Exploring for ICT resource (The relevant) (Hard ware, soft ware) evaluate and adoption of ICT resources

5.2: Pedagogy –Analysis: integrating with ICT and teaching

5.3: Cyber law and protection free software's

5.4: Integrating ICT in Assessment of port folios rubies and data management

5.5: Preparation of multimedia lessons in subjects and planning

5.6: Activities to be conducted in multimedia lessons, the role of the teacher before during and after multimedia lessons, social media and their role in learning

(Twitter, facebook, whatsapp)

5.7: Online learning courses for teachers professional development

5.8: Open education resources, ICT platforms and MOOC

Objectives :

1. Understand different ICT hardware , software resources in the computer kit
2. Use ICT resources in evaluation
3. Determination of various teaching topics and teaching methods and link ICT resources.
4. Get awareness of cycle law, moral issues and security
5. Become aware of the resources ,and teaching needed for scaffolding
6. Prepare multimedia based lessons and make them ready
7. Know of maintenance of portfolios.
8. Understand how, ICT platforms facilitate professional development of teachers.

Introduction:

Achievement level in children improves with ICT based learning .there is a lot of difference between teaching in a traditional classrooms and an of an ICT based class rooms. Installation ICT materials, provision of resources and training are found in this mode multimedia, computer, TV and other ICT device motivate students they reflect reality

(Page 146)

With the integration and use of ICT ,the students ,learn freely according to their abilities ICT resources help is understand the changes taking place in other part of the world ,and modernization of producing of facilitated the learners get knowledge very fast and accordingly to their interests and abilities

5.1 information and communication technology resource:

The ICT resources can be broadly classified two types

1. Hard ware resources
2. Soft ware resources

5.1.1 Hard ware resources:

1. Radio and television:

Radio and television programmes are available at reasonable price they penetrate to all regions and gives information to students they can be used as easily available resources

2. Computer and internet:

Their place in teaching learning process in crucial with the help of these , presentation demonstration and manipulation can be done easily simulations tutorials viotal laboratories visualization and graphical representation can be taken up

3. Interactive boards : smart boards

Smart board can be use as common computer with the help of touch control. There is an interactive board computer, projector in a smart board DVT technology (digital vision touch) and active digester are used in a smart board it is dynamic because of chance of more information and work line coordination.

4. CD & DVD

CD (compact disk) DVD (digital versatile) have more storage capacitated, CDs storage capacity in 700 MB and DVD has 4.7gb capacity

CD s and DVD s are storage devices to store data.

5. Projectors:

Projectors are used to project a film or a video on screen. The projectors are of many kinds.

- a. opaque projectors
- b. overhead projectors
- c. slide projectors
- d. Lcd projectors
- e. DIP (digital light processing) projectors
- f. Digital projectors: this is also called video projectors. With the help of this 3D videos can be shown.

5.1.2 Software Resources

1. Audio Software:

Audio software's are available to record, listen, edit, and mix the tracks. Some software's are available free of cost .some software's can be bought

E.g.: Sony creative software, wave pad audio editing, mix pad etc.

2. Video Software:

Audio software's are used to record; video software's are used to watch videos. Video software's are available in Microsoft, apple etc operating systems. We have to buy advanced video software's, if needed, for example VLC video players, Mx Players, windows movie maker, Adobe Photoshop, free make video downloader etc.

3. Multimedia Software:

Multi media has information about audio, video, and animation. Information to use multimedia software's, multimedia computers are needed. Along with the ordinary computer, camera, scanner, internet, multimedia. Software's are essential a part of multimedia software comes along with operating system (OS). The remaining software we have to upload on the basis of the need. Format factory, adobe flash, blue ray software etc are some examples of this

4. Editing Software:

The software that is useful in altering the audios, videos and graphics with the help of computer is called editing software. E.g.: wave pad, video pad, windows movie maker etc.

5. Web Application Software:

The chat server software application to use web browser is called web application

E.g.: Java, flash web 2.0

With the help of this, so much of information can be obtained fast and comprehensively. It not only helps in getting information but also help helps in sharing it with others. Web technologies help in accessing their peers beyond classrooms.

6. Internet Software:

ICT can be brought to classroom with the help of internet. Internet is the “world wide net work of computers” “with the help of internet connection, information can be transmitted to people having internet connection ,information can be transmitted to people having internet connection within no time , in order to have internet connection ,we should a computer , modern ,telephone ,browsers software , membership with internet provide with the help of internet .

- 1) WWW 2) E-MAIL 3) [FTP](#) 4) [NEWS](#) GROUPS 5) SEARCH ENGINE
- 6) IRC (Inter national relay chat) 7) E-commerce etc.

7. Animation Software:

Projection of all still pictures with minor changes very fast will lead to changes of movements and illusion. This is called to Animation, with the help of analog media and animated jig flash animation digital video formats animation can be recorded.

To project animation digital camera computer and projector etc are used.

8- Simulation Software:

Simulation is the imitation of a real world process. If the simulation is virtual it is not possible to train, test experiment in real life situations simulation is used. Simulation can be used as a representation of reality

5.1.3 Adaption of ICT Resources:

1. Tele collaboration:

The organized use of web resources and allied devices for curricular needs is called “Tele Collaboration”. Tele collaboration is a process of people from different places joint work lines academic purpose with the help of internet devices and resources.

Use ICT is in vogue wherever necessary in school education and we are using them is Tele collaboration Tele conference and video conference and such media

Uses:

1. It keeps and sustains the learners interest, cooperation , creativity and unity
2. E mails and interactive websites help a lot in tele collaboration projects'

2. Tele Conferencing:

The interactive electronic communication between people from two or more places is called Tele conferencing

Tele conferencing facilitates discussions, speeches, interaction with expert and resources persons etc. it is useful in reaching the information to students and clarifying doubts then and there.

5.2 Integration of ICT – Analysis of topics

Different methods are needed to reach different topics in different subjects. The concepts, principals, generalizations, theories etc that are learnt in schools. The ways of reaching these to students are teaching methods.

Language Lessons –Integration of ICT

The lessons are to be selected depending on the cognitive levels of students. Tape recorder, CD player, video player's language lab, and computers with multimedia facilities are needed. Video, audio or other ICT resources should be used in teaching in an appropriate manner.

Mathematics integration of ICT mathematical concepts are to be taught based on the children's knowledge and abilities. Abacus, calculators, scientific calculators, equation solving calculators and ICT devices like computers are very much helpful in the study of mathematics.

Sciences Integration of ICT

Teaching should be done in such a way that cognitive skills and other skills needed by the children are developed. Virtual reality and simulation are helpful in understanding scientific concepts computers, tabs, data base, web cameras online material and other ICT resources are very much useful in reaching science related concepts to children

Social Sciences-Integration of ICT

ICT devices like online maps, database, wikis, blogs, audio, videos are very much helpful in teaching social success meaning full

In this manner ,smart phones ,camera (digital ,hands) smart boards , projector, broad casting ,web pages , internet , virtual tools, wikis , blogs animation and simulation re useful

In teaching social science meaningfully and realistically

Activity: Prepare a power point for one mathematics topic and teach it with the help of multimedia

5.2.1 Alternative means and ICT

Along with hardware and software using alternative resources and means teaching learning process will be better stimulated

Alternative means:

- 1) Wikis
- 2) Blogs
- 3) Pod casts
- 4) Web tools

Construction of knowledge is achieved through integration of ICT and other models, based on skinner's, gagnes cognitive. This knowledge sustain forever the construction approached like John Dewey social activity theory the scaffolding theory proposed by vygotsky, team piaget and Jerome burner's child development theories and gardeners multiple intelligences theory can be integrated with ICT and facilitate construction of knowledge among learners.

5.2.2. ICT RESOURCES AND GADJETS THAT FACILITATE SCAFFOLDING

Gradual withdrawal of external support in solving a problem or conducting an activities and achieving, goal are facilitated by the teaching method scaffolding

Visual technology and virtual reality and real life experience help a lot in promoting scaffolding that was proposed by vygotsky

Scaffolding strategies as suggested by Rebecca abler:

1. Showing all telling
2. Giving to talk
3. Sabha kosam bodana (page 151)
4. Utilization of visual devices
5. Internal –questioning –interval – reviewing

5.2.3 Integration Lesson with ICT classroom management

The class room should be organized on such a way that the children feel that the classroom arranged from them

The planks and chairs and ICT devices should be arranged in such a way that children can comfortable listen watch and participate in the ICT integrated lessons.

Class room ambience should facilitate learning there should not be any discount or abstract to children. Classroom should be safe , favorable and pleasant.

All ICT devices like computer, laptops, smart phones, interactive boards, projectors, TVs

The condition in the classroom alone ICT equipment favorable for children to work either individually or in groups.

Pictures showing the use of ICT equipment, charts, maps and graphs participating to lessons should be kept on walls to facilitate learning.

The projects, assignment given to children should be suitable to the ability of children they should arouse the interest of learners.

5.3 Cyber laws and security; free software's

Now a day's computers and internet have become very popular they are widely also cybercrimes and cheating has increased. In order legally protect people from the crimes; cyber law and security are enforced.

5.3.1 Cyber law

All legal issues in using internet part together are called cyber law . application of activity performed by using computers ,internal and other networks to civil and criminal justice is called cyber law

Think what can be done to control the cheating while using internet

Issues covered in cyber law

1. Internet access
2. Internet usage
3. Freedom of expression
4. Copy right
5. Open source license
6. Data privacy
7. Cyber terrorism
8. Credit card cardious
9. Online gambling

5.3.2 Cyber Security:

Cyber security is the set of practices, technologies and process to protect computers and information from attacking, corrupting and an official usage it is a set of security measures to protect computers, networks, programmers and information from unofficial accessing altering destroying.

Issues covered in cyber security

1. Security to application
2. Security to information
3. Security to discuss
4. Disaster recovery
5. End-user education

Cybercrimes are always novel. So, measures are initiated for cyber security. If adequate measures are not taken there will be server loss to hardware, software and data

Some of the cyber crimes

1. Virus:

This is also a software virus are sent to destroy computer programs files. Virus destroys valuable software and information we have to install firewall, antivirus software etc on our computer

2. Hacking:

The personal information is being stolen and used for illegal activities. We have to install software's in our computers to protect ourselves from hacking. we should not pay attention to unnecessary interaction

3. Plagiarism :

Some people are using information from other online source. They are claiming that the work in their own work this is called plagiarism. In order to prevent this we have to install software like www.plagiarism.com

4. Privacy:

Individual, professional teachers, students maintain personal information in computers stealing information, misusing information and attempting to making profit from the information in order to prevent this AUP (Authorized user policy) needs to be implemented

5. Illegal download /software privacy :

Software's from different companies, films and other videos are downloaded this needs prevention illegal downloads and privacy should be prevented.

5.3.3 Free Software's

The software that are available free of cost are called free software, this called shareware. Free software can be used free of cost. But altering software's or sharing then should not be done without permission of the designer free software cannot to all needs .so, the advanced versions of free software may be bought.

The rimer versions of free software's and the free software's that have advertisement support should be sold and designers will be benefited

e.g. Skype , adobe, acrobat reader, team viewer, sound app, cd burner, Firefox, flash net, u torrent, format factory , winrm ,avast ,avg ,opera,etc are freely available we have to buy some advanced versions like bb edit xn view.

5.4 Integration of ICT in Assessment

We can integrate ICT in classroom Assessment. First we have to assess the features of ICT and use them only when they are useful

5.4.1 Portfolios Assessment

Portfolios are the collection of works /art firms of children /teach in the form of a file. These can be stored and assessed with the help of computer or ICT related

God set s for example , the answer scripts oral drawing etc all can be stored year wise /class wise in the firm of audio , video ppt , scan etc . This facilitates Assessment of Childs progress. It a poem is recited and recovered then the student can listen to the poem and make necessary improvements subsequently he/she can make improvements and try to song systemically some items can be added to portfolios some items can be added by the teachers.

Students make the portfolios and add their work to it. So they will learn in the process of making portfolios.

As part of learning they will notice their mistakes and correct them, students assess on the basis of portfolios. Sometimes they will be made a part of assessment so no time will be wasted for direct assessment this is one learning experience portfolio assessment is the authorized way of assistive children's progress in the long run look 2.8.6

5.4.2 Rubric :

The main purpose of a rubric is to assess children's achievement level or achievements. The souring guide that is made for evaluating constructed response (CR) is called rubric

The Aspects of a Rubric

1. One or more feature to made decision on Childs response.
2. Definitions and examples to explain the meaning of every aspect
3. The needed scoring scale values to dewed the level of each aspect

Steps in the preparation of Rubrics

The steps in the preparation of scoring rubric as suggested by Goodrich

1. Student should be shown different rubric models the children should be given certain devices model assignment so that they can review the assignment

2. The aspects of scoring rubric should be prepared the aspects are taken up for discussion
3. The rubric models should be practiced
4. The efficiency of Arabic needs to be reviewed based on the feedback
5. Children's work should be taken up for assessment

5.4.3 Data management

Controlling process of any collected data is called Data management. The teachers have to store the following school information in computers

Data management –school information: information to be stored in computers

1. Particular of students (class wise)
2. Children's attendance particulars
3. Children's marks particulars(class wise , subject wise)
4. Mid day meal scheme particulars
5. Institutional plan –every year
6. Recording children's progress -student wise
7. Funds sanctioned to school –income and expenditure particulars
8. Social ,cultural programs conducted in school –photos , reports etc

5.5 development of multimedia lessons and planning in schools

First plan should be prepared for developing multimedia lessons. We apply the process to any subjects by implanting the plan

Choose any topics in environmental studies. Collect the multimedia materials needed for teaching the lesson. Discuss in classroom.

The steps followed in planning

1. The skills developed by children
2. The objectives of the lesson
3. Implementation of a multi media lesson
 - Interactive round
 - E pals help (?????) pages (156)
 - Projectors assistance
 - Preparation of PowerPoint
 - Formative assessment should be collected through student response system
4. Extension projects: templates are available online for the development of and presentation multimedia lessons. We can develop multimedia lessons and present them with the help of a computer , internet , interactive white board
Choose a lesson from mathematics prepare a plan based on the steps discussed and teach with the help of multi media

5.6 Activities to be conducted for multimedia lessons

1. **Class room :**

Multimedia computer, projector, interactive white board, audio have to be step up in the class room. Internet connection with proper speed should be ensured

2. **The Teacher :**

The teacher should select a multimedia lesson beforehand. He/she has to analyse the lesson and prepare the slides and script the needed clip art picture, videos, and audios need to be attached. When everything is ready they can present

3. **The Students :**

Multimedia lessons should be developed in an attraction manner to the children the need to have an understanding about the lesson we have to train the students in preparing notes from a power point presentation and responding they should be made ready for involving them in the assessment process held after the presentation.

5.6.1 Multimedia lessons the role of teacher

The teacher should presents slides for the multimedia lessons taken up. In the slides test, video and other effects should be added. Internet, computer, projector, audio, experiment should be arranged. There should enough seating from the teacher and the learner they should be seated face to face to facilitate conversation.

If the students get any kind of doubts, they should be clarified immediately. In the course of session, the understanding of students should be added to the presentation during the session.

After teaching the lesson children should be given assignment based on the lesson. They should be asked to discuss and prepare a report. Apart from that the students understanding should be assessed. If needed the lesson should be taught again.

5.6.2 Social media –Its importance

The science & technology advanced in the whole world. The role computer has gained importance. Then, social media become accessible to all they will help improve the quality in teaching learning.

1. **Social forums :**

Teacher, students and other experts becoming member's social forums and are exchanging opinion. Students are getting information from experts and lectures
e.g : Apetx , AP maths forum, AP subjects forum.

2. Blogs:

The posts that are published on the web are called 'Blogs'. Students will get to know many things from the opinions shared on the blogs. Teaching learning process will be enriched with the help of blogs

e.g: Blogspot, weby websites

3. Twitter:

Tweets are messages up to a maximum 140 characters. These can be read by users who have twitter account. They can also be shared and reposted. Students and teacher an open accounts and there by improve teaching and learning. This is an online social net working media.

4. Face book:

Face book is powerful online social networking services media. We can make profile on face book and invite friends and others users. Opinions, thoughts, and other aspects can be shared by others. Students can share know and develop education information.

5. What's app:

This is online smart phone networking media. We can open an account and share photos, pictures, videos, audios, and information can be shared to group members within seconds. Group chats can be conducted. We can make phone calls also. Teacher and students can use this social medium and improve quality in education matters and learning.

6. Linked In:

This is another important online social networking media. Mostly professionals and people from the field of education are mean less of this forum. All these people will share their opinion.

7. Google plus:

This is a forum for discussion particulars of different courses, many other things about teaching and learning are posted on this forum.

The students and teachers can share their opinions, thoughts innovations can be shared with others. Teaching learning process can be improved.

5.7 Professional Development of Teachers –Online Courses

Teachers take up online course for their professional growth improvement of skills. These courses are offered by some universities in India and institutes from other countries . some universities offered these courses free of cost yet some other charge for the services.

The benefits of online courses:

1. Can continue education

2. Be at home pursue the courses
3. Achieve professional development
4. Choose subject of choice

Online courses in different subjects:

1. Education technology
2. ICT in education
3. Class room, management
4. Assessment
5. Literacy coaching courses
6. Science , technology, engineering and mathematics

As per the provisions of right to education act 2009, the teachers have to continuously develop professionally. In order to achieve this. Teachers can update their skills and knowledge through other distance education mode, apart from online courses. Also workshops, seminars, symposium, conferences, training, orientation programmes can also help the professional growth of teachers.

Wherever needed, the teachers can pursue available online courses in their respective subjects.

5.8 Continuous Professional Development (CPD)

Interactive websites, e-mail, social media, blogs and e-libraries can be used by teaching for their continuous professional development. We can also develop by keeping in touch with others and share information they can make necessary changes to the development and continuous to grow professionally.

Some Relevant Groups:

1. Interactive websites
2. E-mail
3. Social media
4. Blogs
5. Subjects forums

Teacher can grow professionally with the help of groups listed above. They share their opinions, researches, reports and projects with others they get solutions to the problems by posting the problems in the relevant forum. We can share the knowledge in we have to others, which further helps to develop problems. Social media helps to develop professionally. We can share our practices to others and know others opinions. We can get conceptual clarity. Thoughts and practices can be shared with others this makes professional development possible

5.9 ICT Plat forum-MOOC

If we want to rely on non-formal means of education, non-formal courses, open educational sources and open online courses help us we can know more on OER in 43

5.9.1 Various ICT platforms:

“Any basic technology that helps other technologies and process” is called plat form.

In computer language, platform means the operating system that coordinates computer system is called platform. We can notice different resources that are used as ICT platforms.

1. Multimedia computers
2. Laptops
3. Digital cameras
4. Handicam
5. Projectors
6. Television , Radio
7. Smart phones
8. Tablets
9. Interactive boards
10. Social media

5.9.2 Massive open online courses (MOOC)

The courses that are accessible by all and are open to any one are called MOOC. The MOOC s has material to suit to the needs of students and videos, queries ,forums to give answers to questions are organized . Such courses are offered with out and with fee. We call also observe that certain institutes offer courses with the help of IPods, Tablets in the form of software’s. They collect a fixed amount as fee for offering the courses. These don’t have benefits of MODCs. If we want to choose non –formal way of education, non formal courses, open educational resources and online courses help as a lot

The MOOCs are massive open online courses these are offered free of charge. This web based distance learning programme. Large numbers of students from different places can take these courses. These have free courses material along with audio, video, teachings. There are group collaboration, computer based Assessment and feedback are a part of these.

We can join MOOC any time we like and complete it at our own pace there is no limit to the number of takes. We can pursue the courses from wherever we like

It all depends on self pace of learning come credits are given based on the performance and achievement of the takers. As an alternative to MOOCs. Educational software’s are also available on MOOCs

Ministry of human recourses development established SWAYAM (study wets of active learning for young aspiring minds-SWAYAM). The objective of this to offer IIT, IIM,

central universities online courses to Indian citizen quality course material , online lecture ,discussions, assessment quizzes re available to learners through `SWAYAM`

The platform MOOC open ed x is an open source platform offered by ed x org

1. MOOC : Massive open online courses
2. DOCC : Distributed open collaboration courses
3. SPOC : Self-paced online courses

Self Exercise:

- Write about classification of ICT resources
- Explain cyber law and cyber security
- Elaborate the process of integration of ICT in various subjects
- Write about free software
- Write about mapping of assessment tools
- Write about the importance of rubrics
- How is ICT used in port folio assessment?
- Write about assessment of ICT resources and adoption
- What is the role of online course in the professional development of teachers?
- Explain the different models of teachers continuously professional development

Project report / project record

Cover page: Topic, particulars of pupil teacher and guide , particulars of the DIET college

Certificate

Acknowledgement

Chapter-I Introduction

- 1.1.Topic
- 1.2.The need for study
- 1.3.Explanation of the topic / presentation
- 1.4.Limits of the study

Chapter –II Method of Study

- 2.1. Key points of the study
- 2.2.Expected reasons
- 2.3.Expected solutions / activities
- 2.4.Sampling
- 2.5.Data collection tools / implementations of the activity \
- 2.6.Data collection in process
- 2.7.Scoring tools (if necessary)

Chapter –III

3.1. Analysis of study with reasons

3.2. Explanation- graphs or pictures questions of study

Chapter – IV

Conclusions

4.1. Findings

4.2. Suggestions

Appendix

- I. Tools of data collections
- II. Reference books
- III. Abbreviations
- IV. Scoring particulars

Suggested practicum tasks

Activity -1

Discuss the teaching concepts, analysis the concepts

Activity - 2

Preparation of concept maps for the design of content, and concept based curricular material

Activity – 3

Observation classroom discourses, recording, amassing, preparing report and displaying

Activity – 4

Children's literature – perspective other learning material, preparing report and displaying

Activity -5

Select one lesson from five subjects. Select suitable OER multimedia from www and integrated the same with ICT. When you go to schools for the second time prepare lesson plans and teach. Take suggestion from your lecturer make necessary changes to your plan and submit.